

58.2 The Function/Functional Theory

Definitions:

The following function spaces often arise in Fourier analysis. In each case $\mathcal{I}$ is a subinterval of $\mathbb{R}$, and all functions on $\mathcal{I}$ are assumed to be complex valued.

- $\mathcal{C}^0(\mathcal{I})$, the normed linear space of bounded and continuous functions on $\mathcal{I}$ with the norm $\|f\| = \sup\{|f(x)| : x \in \mathcal{I}\}$.
- $\mathcal{L}^1(\mathcal{I})$, the normed linear space of absolutely integrable functions on $\mathcal{I}$ with the norm $\|f\| = \int_{\mathcal{I}} |f(x)| dx$.
- $\mathcal{L}^2(\mathcal{I})$, the inner product space of square integrable functions on $\mathcal{I}$ with the inner product $\langle f, g \rangle = \int_{\mathcal{I}} f(x) \overline{g(x)} dx$.

If $\mathcal{I}$ is not specified, then $\mathcal{I} = \mathbb{R}$.

Let $\phi \in \mathcal{L}^1(\mathbb{R})$. The **Fourier transform** $\mathcal{F}[\phi]$ and the **inverse Fourier transform** $\mathcal{F}^{-1}[\phi]$ of ϕ are the functions

$$\mathcal{F}[\phi]|_x = \int_{-\infty}^{\infty} \phi(y) e^{-i2\pi xy} dy \quad \text{and} \quad \mathcal{F}^{-1}[\phi]|_x = \int_{-\infty}^{\infty} \phi(y) e^{i2\pi xy} dy.$$

Additionally, the terms “Fourier transform” and “inverse Fourier transform” can refer to the processes for computing these functions from ϕ .

A function ϕ on $\mathbb{R}$ is said to be **periodic** if there is a positive value p , called a **period** for ϕ , such that

$$\phi(x + p) = \phi(x) \quad \forall x \in \mathbb{R}.$$

The smallest period, if it exists, is called the **fundamental period**.

The **Fourier series** for a suitably integrable, periodic function ϕ with period $p > 0$ is the infinite series

$$\sum_{k=-\infty}^{\infty} c_k e^{i2\pi \omega_k x}$$

where

$$\omega_k = \frac{k}{p} \quad \text{and} \quad c_k = \frac{1}{p} \int_0^p \phi(y) e^{-i2\pi \omega_k y} dy.$$

The theory for Fourier transforms and series can be generalized so that the requirement of ϕ being “suitably integrable” can be greatly relaxed (see [How01, Chap. 20, 30–34] or [Str94, Chap. 1–4]). Within this generalized theory the **delta function** at $a \in \mathbb{R}$, δ_a , is the functional limit

$$\delta_a(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{2\epsilon} \text{pulse}_{\epsilon}(x - a) \quad \text{where} \quad \text{pulse}_{\epsilon}(s) = \begin{cases} 1 & \text{if } |s| \leq \epsilon \\ 0 & \text{if } |s| > \epsilon \end{cases}.$$

That is, δ_a is the (generalized) function such that

$$\int_{-\infty}^{\infty} \psi(x) \delta_a(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{\infty} \psi(x) \frac{1}{2\epsilon} \text{pulse}_{\epsilon}(x - a) dx$$

whenever ψ is a function continuous at a .

An **array of delta functions** is any expression of the form

$$\sum_{k=-\infty}^{\infty} c_k \delta_{k\Delta x}$$

where $\Delta x > 0$ is fixed (the **spacing** of the array) and $c_k \in \mathbb{C}$ for each $k \in \mathbb{Z}$. If the array is also periodic with period p , then the corresponding **index period** is the positive integer N such that

$$p = N\Delta x \quad \text{and} \quad \phi_{k+N} = \phi_k \quad \forall k \in \mathbb{Z}.$$

The **convolution** $\phi * \psi$ of a suitably integrable pair of functions ϕ and ψ on $\mathbb{R}$ is the function given by

$$\phi * \psi(x) = \int_{-\infty}^{\infty} \phi(x-y)\psi(y) dy.$$

Facts:

All the following facts except those with specific reference can be found in [How01] or [Str94].

1. Warning: Slight variations of the above integral formulas, e.g.,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(y)e^{-ixy} dy \quad \text{and} \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(y)e^{ixy} dy,$$

are also often used to define, respectively, $\mathcal{F}[\phi]$ and $\mathcal{F}^{-1}[\phi]$ (or even $\mathcal{F}^{-1}[\phi]$ and $\mathcal{F}[\phi]$). There is little difference in the resulting “Fourier theories,” but the formulas resulting from using different versions of the Fourier transforms do differ in details. Thus, when computing Fourier transforms using different tables or software, it is important to take into account the specific integral formulas on which the table or software is based.

2. Both $\mathcal{F}$ and $\mathcal{F}^{-1}$ are continuous, linear mappings from $\mathcal{L}^1(\mathbb{R})$ into $\mathcal{C}^0(\mathbb{R})$. Moreover, if both ϕ and $\mathcal{F}[\phi]$ are absolutely integrable, then $\mathcal{F}^{-1}[\mathcal{F}[\phi]] = \phi$.
3. In the generalized theory (which will be assumed hereafter), $\mathcal{F}$ and $\mathcal{F}^{-1}$ are defined on a linear space of (generalized) functions that contains all elements of $\mathcal{L}^1(\mathbb{R})$ and $\mathcal{L}^2(\mathbb{R})$, all Fourier transforms of elements of $\mathcal{L}^1(\mathbb{R})$ and $\mathcal{L}^2(\mathbb{R})$, all piecewise continuous periodic functions, the delta functions, and all periodic arrays of delta functions. $\mathcal{F}$ and $\mathcal{F}^{-1}$ are one-to-one linear mappings from this space onto this space, and are inverses of each other.
4. Every nonconstant, piecewise continuous periodic function has a fundamental period, and every period of such a function is an integral multiple of that fundamental period. Moreover, any two Fourier series for a single periodic function computed using two different periods will be identical after simplification (e.g., after eliminating zero-valued terms).
5. The Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi \omega_k x}$ for a periodic function ϕ with period p can be written in *trigonometric form*

$$c_0 + \sum_{k=1}^{\infty} [a_k \cos(2\pi \omega_k x) + b_k \sin(2\pi \omega_k x)]$$

where

$$a_k = c_k + c_{-k} = \frac{2}{p} \int_0^p \phi(x) \cos(2\pi \omega_k x) dx$$

and

$$b_k = ic_k - ic_{-k} = \frac{2}{p} \int_0^p \phi(x) \sin(2\pi \omega_k x) dx.$$

6. On $\mathcal{I} = (0, p)$ (or any other interval $\mathcal{I}$ of length p), the exponentials

$$\{e^{i2\pi \omega_k x} : \omega_k = k/p, k \in \mathbb{Z}\}$$

form an orthogonal set in $\mathcal{L}^2(\mathcal{I})$, and the Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$ for a periodic function ϕ with period p is simply the expansion of ϕ with respect to this orthogonal set. That is,

$$c_k = \frac{\langle \phi(x), e^{i2\pi\omega_k x} \rangle}{\|e^{i2\pi\omega_k x}\|^2} \quad \forall k \in \mathbb{Z}.$$

Similar comments apply regarding the trigonometric form for the Fourier series and the set

$$\{1, \cos(2\pi\omega_k x), \sin(2\pi\omega_k x) : \omega_k = k/p, k \in \mathbb{N}\}.$$

7. A periodic function ϕ can be identified with its Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$ under a wide range of criteria. In particular:

- If ϕ is smooth, then its Fourier series converges uniformly to ϕ .
- If ϕ is piecewise smooth, then its Fourier series converges pointwise to ϕ everywhere ϕ is continuous.
- If ϕ is square-integrable on $(0, p)$, then

$$\lim_{(M,N) \rightarrow (-\infty, \infty)} \int_0^p \left| \phi(x) - \sum_{k=M}^N c_k e^{i2\pi\omega_k x} \right|^2 dx = 0.$$

- Within the generalized theory,

$$\lim_{(M,N) \rightarrow (-\infty, \infty)} \int_{-\infty}^{\infty} \psi(x) \left[\phi(x) - \sum_{k=M}^N c_k e^{i2\pi\omega_k x} \right] dx = 0$$

whenever ψ is a sufficiently smooth function vanishing sufficiently rapidly at infinity (e.g., any Gaussian function).

8. [BH95, p. 186] Suppose ϕ is a periodic function with Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$. If ϕ is m -times differentiable on $\mathbb{R}$ and $\phi^{(m)}$ is piecewise continuous for some $m \in \mathbb{N}$, then there is a finite constant β such that

$$|c_k| \leq \frac{\beta}{|k|^m} \quad \forall k \in \mathbb{Z}.$$

9. For each $a \in \mathbb{R}$ and function ϕ continuous at a :

$$\int_{-\infty}^{\infty} \phi(x) \delta_a(x) dx = \phi(a) \quad \text{and} \quad \phi \delta_a = \phi(a) \delta_a.$$

10. For each $a \in \mathbb{R}$:

$$\begin{aligned} \mathcal{F}[\delta_a]|_x &= e^{-i2\pi ax} & \text{and} & & \mathcal{F}^{-1}[e^{-i2\pi ax}] &= \delta_a. \\ \mathcal{F}[e^{i2\pi ax}]|_x &= \delta_a & \text{and} & & \mathcal{F}^{-1}[\delta_a]|_x &= e^{i2\pi ax}. \end{aligned}$$

11. A function ϕ is periodic with period p and Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$ if and only if its transforms are the arrays

$$\mathcal{F}[\phi] = \sum_{k=-\infty}^{\infty} c_k \delta_{k\Delta\omega} \quad \text{and} \quad \mathcal{F}^{-1}[\phi] = \sum_{k=-\infty}^{\infty} c_{-k} \delta_{k\Delta\omega}$$

with spacing $\Delta\omega = 1/p$.

12. If

$$\phi = \sum_{k=-\infty}^{\infty} \phi_k \delta_{k\Delta x}$$

is a periodic array with spacing Δx , period p , and corresponding index period N , then:

- The Fourier series for ϕ is given by

$$\sum_{n=-\infty}^{\infty} \Phi_n e^{i2\pi\omega_n x}$$

where $\omega_n = n/p = n/(N\Delta x)$ and

$$\Phi_n = \frac{1}{N\Delta x} \sum_{k=0}^{N-1} \phi_k e^{-i2\pi kn/N}.$$

- The Fourier transforms of ϕ are also periodic arrays of delta functions with index period N . Both have spacing $\Delta\omega = 1/p$ and period $P = 1/\Delta x$. These transforms are given by

$$\mathcal{F}[\phi] = \sum_{n=-\infty}^{\infty} \Phi_n \delta_{n\Delta\omega} \quad \text{and} \quad \mathcal{F}^{-1}[\phi] = \sum_{n=-\infty}^{\infty} \Phi_{-n} \delta_{n\Delta\omega}.$$

13. (*Convolution identities*) Let f, F, g , and G be functions with $F = \mathcal{F}[f]$ and $G = \mathcal{F}[g]$. Then, provided the convolutions exist,

$$\mathcal{F}[fg] = F * G \quad \text{and} \quad \mathcal{F}[f * g] = FG.$$

Examples:

1. For $\alpha > 0$,

$$\begin{aligned} \mathcal{F}[\text{pulse}_\alpha]|_x &= \int_{-\infty}^{\infty} \text{pulse}_\alpha(y) e^{-i2\pi xy} dy \\ &= \int_{-\alpha}^{\alpha} e^{-i2\pi xy} dy \\ &= \frac{e^{i2\pi\alpha x} - e^{-i2\pi\alpha x}}{i2\pi x} = \frac{\sin(2\pi\alpha x)}{\pi x}. \end{aligned}$$

2. Let f be the periodic function,

$$f(x) = \begin{cases} |x| & \text{for } -1 \leq x < 1 \\ f(x-2) & \text{for all } x \in \mathbb{R} \end{cases}.$$

The period of this function is $p = 2$, and its Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$ is given by

$$\omega_k = \frac{k}{2} \quad (\text{so } e^{i2\pi\omega_k x} = e^{i\pi kx})$$

and

$$\begin{aligned} c_k &= \frac{1}{2} \int_0^2 \begin{cases} |x| & \text{if } x < 1 \\ |x-2| & \text{if } 1 < x \end{cases} e^{-i\pi kx} dx \\ &= \begin{cases} 1/2 & \text{if } k = 0 \\ (k\pi)^{-2} [(-1)^k - 1] & \text{if } k \neq 0 \end{cases}. \end{aligned}$$

Since f is continuous and piecewise smooth, f equals its Fourier series,

$$f(x) = \frac{1}{2} + \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{(-1)^k - 1}{k^2 \pi^2} e^{i\pi k x} \quad \forall x \in \mathbb{R}.$$

Moreover,

$$\mathcal{F}[f] = \frac{1}{2} \delta_0 + \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{(-1)^k - 1}{k^2 \pi^2} \delta_{k/2}.$$

3. Let

$$\phi = \sum_{k=-\infty}^{\infty} \phi_k \delta_{\Delta x}$$

be the periodic array with spacing $\Delta x = 1/3$, index period $N = 4$, and coefficients

$$\phi_0 = 0, \quad \phi_1 = 1, \quad \phi_2 = 2, \quad \text{and} \quad \phi_3 = 1.$$

The period p of ϕ is then

$$p = N \Delta x = 4 \cdot \frac{1}{3} = \frac{4}{3}.$$

Its Fourier transform $\Phi = \mathcal{F}[\phi]$ is also a periodic array,

$$\Phi = \sum_{n=-\infty}^{\infty} \Phi_n \delta_{n \Delta \omega},$$

with index period $N = 4$. The spacing $\Delta \omega$ and period P of Φ are determined, respectively, from the period p and spacing Δx of ϕ by

$$\Delta \omega = \frac{1}{p} = \frac{1}{4/3} = \frac{3}{4} \quad \text{and} \quad P = \frac{1}{\Delta x} = \frac{1}{1/3} = 3.$$

The coefficients are given by

$$\begin{aligned} \Phi_n &= \frac{1}{N \Delta x} \sum_{k=0}^{N-1} \phi_k e^{-i2\pi k n / N} \\ &= \frac{1}{N \Delta x} [\phi_0 e^{-i2\pi 0 n / 4} + \phi_1 e^{-i2\pi 1 n / 4} + \phi_2 e^{-i2\pi 2 n / 4} + \phi_3 e^{-i2\pi 3 n / 4}] \\ &= \frac{1}{4(1/3)} [0e^0 + 1e^{-in\pi/2} + 2e^{-in\pi} + 1e^{-i3n\pi/2}] \\ &= \frac{3}{4} [0 + (-i)^n + 2 + i^n]. \end{aligned}$$

Thus,

$$\Phi_0 = 3, \quad \Phi_1 = \frac{3}{2}, \quad \Phi_2 = 0 \quad \text{and} \quad \Phi_3 = \frac{3}{2}.$$

Applications:

1. [BC01, p. 155] or [Col88, pp. 159–160] (*Partial differential equations*) Using polar coordinates, the steady-state temperature u at position (r, θ) on a uniform, insulated disk of radius 1 satisfies

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0 \quad \text{for } 0 < r < 1.$$

As a function of θ , u is periodic with period 2π and has (equivalent) Fourier series

$$\sum_{k=-\infty}^{\infty} c_k e^{ikx} \quad \text{and} \quad c_0 + \sum_{k=1}^{\infty} [a_k \cos(kx) + b_k \sin(kx)]$$

where the coefficients are functions of r . If u satisfies the boundary condition

$$u(1, \theta) = f(\theta) \quad \text{for } 0 \leq \theta < 2\pi$$

for some function f on $[0, 2\pi)$, then the coefficients in the series can be determined, yielding

$$u(r, \theta) = \sum_{k=-\infty}^{\infty} r^k \gamma_k e^{ik\theta} = \gamma_0 + \sum_{k=1}^{\infty} [r^k \alpha_k \cos(k\theta) + r^k \beta_k \sin(k\theta)]$$

where

$$\gamma_k = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-ik\theta} d\theta,$$

$$\alpha_k = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos(k\theta) d\theta, \quad \text{and} \quad \beta_k = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin(k\theta) d\theta.$$

2. (*Systems analysis*) In systems analysis, a “system” S transforms any “input” function f_I to a corresponding “output” function $f_O = S[f_I]$. Often, the output of S can be described by the convolution formula

$$f_O = h * f_I$$

where h is some fixed function called the impulse response of the system. The corresponding Fourier transform $H = \mathcal{F}[h]$ is the system’s transfer function. By the convolution identity, the output is also given by

$$f_O = \mathcal{F}^{-1}[HF_I] \quad \text{where } F_I = \mathcal{F}[f_I].$$

Two such systems are

- A delayed output system, for which

$$h(x) = \delta_T(x) \quad \text{and} \quad H(\omega) = e^{-i2\pi T\omega}$$

for some $T > 0$. Then

$$f_O(x) = f_I * \delta_T(x) = \int_{-\infty}^{\infty} f_I(x-y) \delta_T(y) dy = f_I(x-T).$$

- [ZTF98, p. 178] or [How01, p. 477] An ideal low pass filter, for which

$$h(x) = \frac{\sin(2\pi\Omega x)}{\pi x} \quad \text{and} \quad H(y) = \text{pulse}_{\Omega}(y)$$

for some $\Omega > 0$. If an input function f_I is periodic with Fourier series $\sum_{k \in \mathbb{Z}} c_k e^{i2\pi\omega_k x}$, then

$$\begin{aligned} HF_I &= \text{pulse}_\Omega \cdot \sum_{k=-\infty}^{\infty} c_k \delta_{\omega_k} \\ &= \sum_{k=-\infty}^{\infty} c_k \text{pulse}_\Omega(\omega_k) \delta_{\omega_k} \\ &= \sum_{k=-\infty}^{\infty} c_k \begin{cases} 1 & \text{if } |\omega_k| \leq \Omega \\ 0 & \text{if } |\omega_k| > \Omega \end{cases} \delta_{\omega_k} = \sum_{|\omega_k| \leq \Omega} c_k \delta_{\omega_k}. \end{aligned}$$

Thus,

$$f_O(x) = \mathcal{F}^{-1}[HF_I]|_x = \mathcal{F}^{-1} \left[\sum_{|\omega_k| \leq \Omega} c_k \delta_{\omega_k} \right] \Big|_x = \sum_{|\omega_k| \leq \Omega} c_k e^{i2\pi\omega_k x}.$$

3. [ZTF98, pp. 520–521] (*Deconvolution*) Suppose S is a system given by

$$S[f_I] = h * f_I,$$

but with h and $H = \mathcal{F}[h]$ being unknown. Since

$$f_O = \mathcal{F}^{-1}[HF_I] \quad \text{where } F_I = \mathcal{F}[f_I],$$

both h and H can be determined as follows:

- Find the output $f_O = S[f_I]$ for some known input f_I for which $F_I = \mathcal{F}[f_I]$ is never zero.
- Compute

$$H = \frac{F_O}{F_I} \quad \text{where } F_O = \mathcal{F}[f_O].$$

- Compute $h = \mathcal{F}^{-1}[H]$.

Similarly, an input f_I can be reconstructed from an output f_O by

$$f_I = \mathcal{F}^{-1} \left[\frac{F_O}{H} \right]$$

provided the transfer function H is known and is never zero.

58.3 The Discrete Theory

Definitions:

In all the following, $N \in \mathbb{N}$, and the indexing of any “ N items” (including rows and columns of matrices) will run from 0 to $N - 1$.

An N th **order sequence** is an ordered list of N complex numbers,

$$(c_0, c_1, c_2, \dots, c_{N-1}).$$

Such a sequence will often be written as the column vector

$$\mathbf{c} = [c_0, c_1, c_2, \dots, c_{N-1}]^T,$$

and the k th component of the sequence will be denoted by either c_k or $[\mathbf{c}]_k$. In addition, any such sequence will be viewed as part of an infinite repeating sequence

$$(\dots, c_{-1}, c_0, c_1, c_2, \dots, c_{N-1}, c_N, \dots)$$

in which $c_{N+k} = c_k$ for all $k \in \mathbb{Z}$.

Let $\mathbf{c}$ be an N th order sequence. The (N th order) **discrete Fourier transform (DFT)** $\widehat{\mathcal{F}}_N \mathbf{c}$ and the (N th order) **inverse discrete Fourier transform (inverse DFT)** $\widehat{\mathcal{F}}_N^{-1} \mathbf{c}$ of $\mathbf{c}$ are the two N th order sequences given by

$$[\widehat{\mathcal{F}}_N \mathbf{c}]_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} c_k e^{-i2\pi kn/N} \quad \text{and} \quad [\widehat{\mathcal{F}}_N^{-1} \mathbf{c}]_k = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} c_n e^{i2\pi nk/N}.$$

Additionally, the terms “discrete Fourier transform” and “inverse discrete Fourier transform” can refer to the processes for computing these sequences from $\mathbf{c}$.

Given two N th order sequences $\mathbf{a}$ and $\mathbf{b}$, the corresponding product $\mathbf{a}\mathbf{b}$ and convolution $\mathbf{a} * \mathbf{b}$ are the N th order sequences given by

$$[\mathbf{a}\mathbf{b}]_k = a_k b_k \quad \text{and} \quad [\mathbf{a} * \mathbf{b}]_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} a_{k-j} b_j.$$

Facts:

All the following facts can be found in [BH95, Chap. 2, 3] or [How01, Chap. 38, 39].

1. Warning: Slight variations of the above summation formulas, e.g.,

$$\sum_{k=0}^{N-1} c_k e^{-i2\pi kn/N} \quad \text{and} \quad \frac{1}{N} \sum_{n=0}^{N-1} c_n e^{i2\pi nk/N},$$

are also often used to define, respectively, $\widehat{\mathcal{F}}_N \mathbf{c}$ and $\widehat{\mathcal{F}}_N^{-1} \mathbf{c}$ (or even $\widehat{\mathcal{F}}_N^{-1} \mathbf{c}$ and $\widehat{\mathcal{F}}_N \mathbf{c}$).

2. $\widehat{\mathcal{F}}_N$ and $\widehat{\mathcal{F}}_N^{-1}$ are one-to-one linear transformations from $\mathbb{C}^N$ onto $\mathbb{C}^N$. They preserve the standard inner product and are inverses of each other.
3. Letting $w = e^{-i2\pi/N}$, the matrices (with respect to the standard basis for $\mathbb{C}^N$) for the N th order discrete Fourier transforms (also denoted by $\widehat{\mathcal{F}}_N$ and $\widehat{\mathcal{F}}_N^{-1}$) are

$$\widehat{\mathcal{F}}_N = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & w^{1 \cdot 1} & w^{1 \cdot 2} & w^{1 \cdot 3} & \dots & w^{1 \cdot (N-1)} \\ 1 & w^{2 \cdot 1} & w^{2 \cdot 2} & w^{2 \cdot 3} & \dots & w^{2 \cdot (N-1)} \\ 1 & w^{3 \cdot 1} & w^{3 \cdot 2} & w^{3 \cdot 3} & \dots & w^{3 \cdot (N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{(N-1)1} & w^{(N-1)2} & w^{(N-1)3} & \dots & w^{(N-1)(N-1)} \end{bmatrix}$$

and

$$\widehat{\mathcal{F}}_N^{-1} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & w^{-1 \cdot 1} & w^{-1 \cdot 2} & w^{-1 \cdot 3} & \cdots & w^{-1(N-1)} \\ 1 & w^{-2 \cdot 1} & w^{-2 \cdot 2} & w^{-2 \cdot 3} & \cdots & w^{-2(N-1)} \\ 1 & w^{-3 \cdot 1} & w^{-3 \cdot 2} & w^{-3 \cdot 3} & \cdots & w^{-3(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{-(N-1)1} & w^{-(N-1)2} & w^{-(N-1)3} & \cdots & w^{-(N-1)(N-1)} \end{bmatrix}.$$

4. $\widehat{\mathcal{F}}_N$ and $\widehat{\mathcal{F}}_N^{-1}$ are symmetric matrices, and are complex conjugates of each other (i.e., $[\widehat{\mathcal{F}}_N]^* = \widehat{\mathcal{F}}_N^{-1}$).
5. (*Convolution identities*) Let $\mathbf{v}$, $\mathbf{V}$, $\mathbf{w}$, and $\mathbf{W}$ be N th order sequences with $\mathbf{V} = \widehat{\mathcal{F}}_N \mathbf{v}$ and $\mathbf{W} = \widehat{\mathcal{F}}_N \mathbf{w}$. Then

$$\widehat{\mathcal{F}}_N[\mathbf{vw}] = \mathbf{V} * \mathbf{W} \quad \text{and} \quad \widehat{\mathcal{F}}_N[\mathbf{v} * \mathbf{w}] = \mathbf{VW}.$$

Examples:

1. The matrices for the four lowest order DFTs are $\widehat{\mathcal{F}}_1 = [1]$,

$$\widehat{\mathcal{F}}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \widehat{\mathcal{F}}_3 = \frac{1}{2\sqrt{3}} \begin{bmatrix} 2 & 2 & 2 \\ 2 & -1 - \sqrt{3} & -1 + \sqrt{3} \\ 2 & -1 + \sqrt{3} & -1 - \sqrt{3} \end{bmatrix},$$

and

$$\widehat{\mathcal{F}}_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & 1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}.$$

2. The DFT of $\mathbf{v} = [1, 2, 3, 4]^T$ is $\mathbf{V} = [V_0, V_1, V_2, V_3]^T$ where

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \\ V_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & 1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 + i \\ -1 \\ -1 - i \end{bmatrix}.$$

Applications:

1. [CG00, p. 153] Assume

$$f(z) = \sum_{n=0}^{2M} f_n z^n$$

is the product of two known M th order polynomials

$$g(z) = \sum_{k=0}^M g_k z^k \quad \text{and} \quad h(z) = \sum_{k=0}^M h_k z^k.$$

The coefficients of f form a sequence $\mathbf{f} = [f_0, f_1, f_2, \dots, f_{2M}]^T$ of order $N = 2M + 1$. These coefficients can be computed from the coefficients of g and h using either of the following approaches:

- Let $\mathbf{g} = [g_0, g_1, g_2, \dots, g_{2M}]^T$ and $\mathbf{h} = [h_0, h_1, h_2, \dots, h_{2M}]^T$ be the N th order sequences in which g_k and h_k are the corresponding coefficients of the polynomials g and h when $k \leq M$ and are zero when $M < k \leq N$. The coefficients of f can then be computed by

$$f_k = \sum_{j=0}^{2M} g_{k-j} h_j = \sqrt{N} [\mathbf{g} * \mathbf{h}]_k \quad \text{for } k = 0, 1, 2, \dots, 2M.$$

- Let $\mathbf{F} = [F_0, F_1, F_2, \dots, F_{2M}]^T$ be the N th order DFT of $\mathbf{f}$, and note that, for $n = 0, 1, 2, \dots, N - 1$,

$$F_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f_k e^{-i2\pi kn/N} = \frac{1}{\sqrt{N}} f(w^n) = \frac{1}{\sqrt{N}} h(w^n) g(w^n)$$

where $w = e^{-i2\pi/N}$. Thus, the coefficients of f can be computed by first using the formulas for g and h to compute

$$F_n = \frac{1}{\sqrt{N}} h(w^n) g(w^n) \quad \text{for } n = 0, 1, 2, \dots, N - 1,$$

and then taking the inverse DFT of $\mathbf{F} = [F_0, F_1, F_2, \dots, F_{N-1}]^T$.

2. [BH95, p.247] Consider finding the solution $\mathbf{v} = [v_0, v_1, v_2, \dots, v_{N-1}]^T$ to the difference equation

$$\alpha v_{k-1} + \beta v_k + \alpha v_{k+1} = f_k \quad \text{for } n = 0, 1, 2, \dots, N - 1$$

where α and β are constants and $\mathbf{f} = [f_0, f_1, f_2, \dots, f_{N-1}]^T$ is a known N th order sequence. If the boundary conditions are periodic (i.e., $v_0 = v_N$ and $v_{-1} = v_{N-1}$), then setting

$$\mathbf{v} = \hat{\mathcal{F}}_N^{-1} \mathbf{V} \quad \text{and} \quad \mathbf{f} = \hat{\mathcal{F}}_N^{-1} \mathbf{F}$$

yields the sequence $\mathbf{v}$ that satisfies the periodic boundary conditions, and whose DFT, $\mathbf{V}$, satisfies

$$\sum_{n=0}^{N-1} C_n e^{i2\pi nk/N} = 0 \quad \text{for } k = 0, 1, 2, \dots, N - 1$$

where

$$C_n = \left[2\alpha \cos\left(\frac{2\pi n}{N}\right) + \beta \right] V_n - F_n.$$

From this, it follows that the solution $\mathbf{v}$ to the original difference equation is the DFT of the sequence $[V_0, V_1, V_2, \dots, V_{N-1}]^T$ given by

$$V_n = \frac{F_n}{2\alpha \cos\left(\frac{2\pi n}{N}\right) + \beta},$$

provided the denominator never vanishes.

3. A difference equation of the form just considered with periodic boundary conditions arises when considering the steady-state temperature distribution on a uniform ring containing heat sources and sinks. In this case, the temperature v , as a function of angular position θ , is modeled by the

one-dimensional Poisson's equation

$$\frac{d^2 v}{d\theta^2} = f(\theta),$$

where $f(\theta)$ describes the heat source/sink density at angular position θ . The discrete analog of this equation is

$$v_{k-1} - 2v_k + v_{k+1} = f_k \quad \text{for } n = 0, 1, 2, \dots, N-1$$

where

$$\mathbf{v} = [v_0, v_1, v_2, \dots, v_{N-1}]^T \quad \text{and} \quad \mathbf{f} = [f_0, f_1, f_2, \dots, f_{N-1}]^T$$

describe, respectively, the temperatures at N evenly space positions around the ring, and the net sources and sinks of thermal energy about these positions. Setting

$$\mathbf{v} = \hat{\mathcal{F}}_N^{-1} \mathbf{V} \quad \text{and} \quad \mathbf{f} = \hat{\mathcal{F}}_N^{-1} \mathbf{F}$$

and applying the formulas given above yields

$$\left[2 \cos \left(\frac{2\pi n}{N} \right) - 2 \right] V_n = F_n \quad \text{for } n = 0, 1, 2, \dots, N-1.$$

The coefficient on the left is nonzero when $n \neq 0$. Thus,

$$V_n = \frac{F_n}{2 \cos \left(\frac{2\pi n}{N} \right) - 2} \quad \text{for } n = 1, 2, \dots, N-1.$$

However, for $n = 0$,

$$0 \cdot V_0 = F_0 = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f_k e^{-i2\pi k \cdot 0/N} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f_k,$$

pointing out that, for an equilibrium temperature distribution to exist, the net applied heat energy, $\sum_{k=0}^{N-1} f_k$, must be zero. Assuming this, the last equation then implies that V_0 is arbitrary, which, since

$$V_0 = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} v_k e^{-i2\pi k \cdot 0/N} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} v_k$$

means that the given conditions are not sufficient to determine the average temperature throughout the ring.

58.4 Relating the Functional and Discrete Theories

Definitions:

Let f be a (continuous) function on a finite interval $[0, L]$. For any $N \in \mathbb{N}$ and $\Delta x > 0$ satisfying $N\Delta x \leq L$, the corresponding (**N th order**) **sampling** (with **spacing** Δx) is the sequence

$$\mathbf{f} = [f_0, f_1, f_2, \dots, f_{N-1}]^T \quad \text{where } f_k = f(k\Delta x).$$

Corresponding to this is the **scaled sampling**

$$\hat{\mathbf{f}} = [\hat{f}_0, \hat{f}_1, \hat{f}_2, \dots, \hat{f}_{N-1}]^T$$

and the **discrete approximation**

$$\hat{f} = \sum_{k=-\infty}^{\infty} \hat{f}_k \delta_{k\Delta x}$$

where, in both,

$$\hat{f}_k = \begin{cases} f_k \Delta x & \text{for } k = 0, 1, 2, \dots, N-1 \\ \hat{f}_{k+N} & \text{in general} \end{cases}.$$

Facts:

1. [How01, pp. 713–715] Let $\hat{f}$ be the discrete approximation of a continuous function f based on an N th order sampling with spacing Δx . Then $\hat{f}$ is a periodic array of delta functions with spacing Δx and index period N that approximates f over the interval $(0, N\Delta x)$. In particular, for any other function ψ which is continuous on $[0, N\Delta x]$,

$$\int_0^{N\Delta x} \psi(x) f(x) dx \approx \int_0^{N\Delta x} \psi(x) \hat{f}(x) dx = \sum_{k=0}^{N-1} \psi(k\Delta x) \hat{f}_k.$$

Attempting to approximate f with $\hat{f}$ outside the interval $(0, N\Delta x)$, however, cannot be justified unless f is also periodic with period $N\Delta x$.

2. [How01, chap. 38] Let

$$\mathbf{f} = [f_0, f_1, f_2, \dots, f_{N-1}]^T \quad \text{and} \quad \mathbf{F} = [F_0, F_1, F_2, \dots, F_{N-1}]^T$$

be two N th order sequences, and let

$$f = \sum_{k=-\infty}^{\infty} f_k \delta_{k\Delta x} \quad \text{and} \quad F = \sum_{n=-\infty}^{\infty} F_n \delta_{n\Delta\omega}$$

be two corresponding periodic arrays of delta functions with index period N , and with spacings Δx and $\Delta\omega$ satisfying $\Delta x \Delta\omega = 1/N$. Then

$$F = \mathcal{F}[f] \iff \mathbf{F} = \Delta\omega \sqrt{N} \hat{\mathcal{F}}_N \mathbf{f}.$$

In particular, if $\Delta\omega = \Delta x = N^{-1/2}$, then

$$F = \mathcal{F}[f] \iff \mathbf{F} = \hat{\mathcal{F}}_N \mathbf{f}.$$

3. [How01, pp. 719–723] Suppose f is a continuous, piecewise smooth function that vanishes outside the finite interval $(0, L)$. Let $F = \mathcal{F}[f]$, and let $\hat{\mathbf{f}}$ be the N th order scaled sampling of f with spacing Δx chosen so that $L = (N-1)\Delta x$. Then, for each $n \in \mathbb{Z}$,

$$F(n\Delta\omega) \approx \sqrt{N} [\hat{\mathcal{F}}_N \hat{\mathbf{f}}]_n$$

where $\Delta\omega = (N\Delta x)^{-1}$. The error in this approximation is bounded by

$$(\max |f'| + 2\pi |n| \Delta\omega \max |f|) \frac{L^2 N}{2(N-1)^2}.$$

In practice, this bound may significantly overestimate the actual error.

4. [BH95, pp. 181–188] If f is a continuous, piecewise smooth, periodic function with period p and Fourier series $\sum_{n \in \mathbb{Z}} c_n e^{i2\pi\omega_n x}$, then

$$c_n \approx \frac{1}{\sqrt{N}} [\widehat{\mathcal{F}}_N \mathbf{f}]_n \quad \text{for } -\frac{N}{2} < n \leq \frac{N}{2}$$

where $\mathbf{f}$ is the N th order sampling of f over $[0, p]$ with spacing $\Delta x = p/N$. Moreover, if f is m -times differentiable on $\mathbb{R}$ and $f^{(m)}$ is piecewise continuous and piecewise monotone for some $m \in \mathbb{N}$, then the error in this approximation is bounded by

$$\frac{\beta}{N^{m+1}}$$

for some positive constant β independent of n and N .

Examples:

1. Let

$$f = \sum_{k=-\infty}^{\infty} f_k \delta_{k/3}$$

be the periodic array with index period $N = 4$ and coefficients

$$f_0 = 0, \quad f_1 = 1, \quad f_2 = 2, \quad \text{and} \quad f_3 = 1.$$

The Fourier transform of f is then the periodic array

$$F = \sum_{n=-\infty}^{\infty} F_n \delta_{n\Delta\omega}$$

with index period 4, spacing $\Delta\omega = 3/4$, and coefficients given by

$$\begin{bmatrix} F_0 \\ F_1 \\ F_2 \\ F_3 \end{bmatrix} = \Delta\omega \sqrt{4} \widehat{\mathcal{F}}_4 \mathbf{f} = \frac{3}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & 1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 \\ 3 \\ 0 \\ -3 \end{bmatrix}.$$

2. The 8th order sampling with spacing $\Delta x = 1/6$ of

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 1/2 \\ 1 - x & \text{for } 1/2 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is $\mathbf{f} = [f_0, f_1, f_2, \dots, f_7]^T$ with

$$f_k = f\left(\frac{k}{6}\right) = \begin{cases} k/6 & \text{for } k = 0, 1, 2, 3 \\ 1 - k/6 & \text{for } k = 4, 5, 6 \\ 0 & \text{for } k = 7 \end{cases}.$$

That is,

$$\mathbf{f} = \left[0, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{2}{6}, \frac{1}{6}, 0, 0\right]^T.$$

Multiplying this by the spacing $\Delta x = 1/6$ yields the scaled sampling

$$\widehat{\mathbf{f}} = \left[0, \frac{1}{36}, \frac{2}{36}, \frac{3}{36}, \frac{2}{36}, \frac{1}{36}, 0, 0 \right]^T.$$

Letting $F = \mathcal{F}[f]$, $\Delta\omega = (N\Delta x)^{-1} = 3/4$, and $w = e^{-i2\pi/8}$,

$$\begin{aligned} F(n\Delta\omega) &\approx \sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_n \\ &= \frac{1}{36} [0w^{0n} + 1w^{1n} + 2w^{2n} + 3w^{3n} + 2w^{4n} + 1w^{5n} + 0w^{6n} + 0w^{7n}] \\ &= \frac{1}{36} w^{3n} [w^{-2n} + 2w^{-n} + 3 + 2w^n + w^{2n}], \end{aligned}$$

which simplifies to

$$F(n\Delta\omega) \approx \sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_n = \frac{1}{36} e^{-i3n\pi/4} \left[3 + 2 \cos\left(\frac{n\pi}{2}\right) + 4 \cos\left(\frac{n\pi}{4}\right) \right].$$

The error bound on this approximation is

$$E_n = (\max |f'| + 2\pi |n| \Delta\omega \max |f|) \frac{L^2 N}{2(N-1)^2} = \frac{4 + 3\pi |n|}{36}.$$

In fact, the Fourier transform of f is easily found to be

$$F(\omega) = e^{-i\pi\omega} \left[\frac{\sin(\pi\omega/2)}{\pi\omega} \right]^2.$$

So

$$F(n\Delta\omega) = F\left(\frac{3n}{4}\right) = e^{-i3n\pi/4} \left[\frac{4 \sin(3n\pi/8)}{3n\pi} \right]^2,$$

and the actual error in the approximation is

$$\begin{aligned} \varepsilon_n &= \left| F(n\Delta\omega) - \sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_n \right| \\ &= \frac{1}{36} \left| 64 \left[\frac{\sin(n\pi/8)}{n\pi} \right]^2 - 3 - 2 \cos\left(\frac{n\pi}{2}\right) - 4 \cos\left(\frac{n\pi}{4}\right) \right|. \end{aligned}$$

For $n = 0$ through 4, the actual and approximate values of $F(n\Delta\omega)$ (to five decimal places) are

$F(0\Delta\omega) = 0.25000$	$\sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_0 = 0.25000$
$F(1\Delta\omega) = -0.10872(1+i)$	$\sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_1 = -0.11448(1+i)$
$F(2\Delta\omega) = 0.02252i$	$\sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_2 = 0.02778i$
$F(3\Delta\omega) = 0.00207(1-i)$	$\sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_3 = 0.00337(1-i)$
$F(4\Delta\omega) = -0.01126$	$\sqrt{8} [\widehat{\mathcal{F}}_8 \widehat{\mathbf{f}}]_4 = -0.02778$

The corresponding error bounds and actual errors (to five decimal places) are

$$\begin{array}{ll} E_0 = 0.11111 & \varepsilon_0 = 0.00000 \\ E_1 = 0.37291 & \varepsilon_1 = 0.00815 \\ E_2 = 0.63471 & \varepsilon_2 = 0.00526 \\ E_3 = 0.89651 & \varepsilon_3 = 0.00183 \\ E_4 = 1.15831 & \varepsilon_4 = 0.01652. \end{array}$$

Applications:

1. Let f be a continuous, piecewise smooth function to be sampled that is nonzero only in the interval $(0, 1)$ and which satisfies

$$|f(x)| \leq \frac{1}{4} \quad \text{and} \quad |f'(x)| \leq 1 \quad \forall x \in \mathbb{R}.$$

Consider the problem of approximating $F(n\Delta\omega)$ where $F = \mathcal{F}[f]$, $\Delta\omega = 1/2$, and n is any integer from 0 to 10. For each $N \in \mathbb{N}$, set

$$\Delta x = \frac{1}{N\Delta\omega} = \frac{2}{N} \quad \text{and} \quad L = (N-1)\Delta x = \frac{2(N-1)}{N},$$

and let $\hat{\mathbf{f}} = [\hat{f}_0, \hat{f}_1, \hat{f}_2, \dots, \hat{f}_{N-1}]^T$ be the corresponding scaled sampling of f ,

$$\hat{f}_k = f(k\Delta x)\Delta x = f\left(\frac{2k}{N}\right) \cdot \frac{2}{N}.$$

Then, for $n = 0, 1, 2, \dots, 10$,

$$F(n\Delta\omega) \approx \sqrt{N} [\hat{\mathcal{F}}_N \hat{\mathbf{f}}]_n = \frac{2}{N} \sum_{k=0}^{N-1} f\left(\frac{2k}{N}\right) e^{-i2\pi kn/N}$$

with an error bound of

$$(\max |f'| + 2\pi |n| \Delta\omega \max |f|) \frac{L^2 N}{2(N-1)^2} = \frac{4 + 10\pi}{2N} < \frac{18}{N}.$$

2. [CG00, p. 157] (*Deconvolution*) Assume $g = f * h$ where g and h are known by N th order samplings $\mathbf{g}$ and $\mathbf{h}$, each with spacing Δx . The deconvolution formula for finding f ,

$$f = \mathcal{F}^{-1}[V] \quad \text{where } V = \frac{G}{H} = \frac{\mathcal{F}[g]}{\mathcal{F}[h]},$$

is approximated by

$$f(k\Delta x) \approx \sqrt{N} \left[\hat{\mathcal{F}}_N^{-1} \hat{\mathbf{V}} \right]_k$$

where, letting $\Delta\omega = (N\Delta x)^{-1}$, $\hat{\mathbf{V}}$ is the sequence given by

$$\hat{V}_n = \Delta\omega \frac{\sqrt{N} \left[\hat{\mathcal{F}}_N[\hat{\mathbf{g}}] \right]_n}{\sqrt{N} \left[\hat{\mathcal{F}}_N[\hat{\mathbf{h}}] \right]_n} \quad \text{for } n = 0, 1, 2, \dots, N-1.$$

This reduces to

$$f(k\Delta x) \approx \frac{1}{\Delta x \sqrt{N}} \left[\widehat{\mathcal{F}}_N^{-1} \mathbf{V} \right]_k$$

where, letting $\mathbf{G} = \widehat{\mathcal{F}}_N \mathbf{g}$ and $\mathbf{H} = \widehat{\mathcal{F}}_N \mathbf{h}$,

$$V_n = \frac{G_n}{H_n} \quad \text{for } n = 0, 1, 2, \dots, N-1.$$

This requires, of course, that each H_n be nonzero.

58.5 The Fast Fourier Transform

A **fast Fourier transform (FFT)** is any of a number of algorithms for computing DFTs in which the number of arithmetic computations is greatly reduced through clever use of symmetries and cyclic structures in the DFT matrices. The FFT described here is the standard *radix 2* FFT for computing the N th order “alternative” discrete Fourier transform $\mathcal{D}_N \mathbf{v}$ given by

$$[\mathcal{D}_N \mathbf{v}]_n = \sqrt{N} [\widehat{\mathcal{F}}_N \mathbf{v}]_n = \sum_{k=0}^{N-1} v_k e^{-i2\pi nk/N}.$$

For additional details on implementing a radix 2 FFT and descriptions of other FFTs, see [BH95, chap. 10], [CG00], or [Wal96].

Algorithms:

Two algorithms for computing the alternative DFT

$$\mathbf{V} = [V_0, V_1, V_2, \dots, V_{N-1}]^T$$

of an N th order sequence

$$\mathbf{v} = [v_0, v_1, v_2, \dots, v_{N-1}]^T$$

are given. The first is the “first level” radix 2 FFT illustrating the basic concepts. The second is the more complete radix 2 FFT.

1. (*First level radix 2 FFT*) This requires $N = 2M$ for some $M \in \mathbb{N}$:

- First, split $\mathbf{v}$ into two M th order sequences $\mathbf{v}^O$ and $\mathbf{v}^E$ composed, respectively, of the even-indexed and odd-indexed elements of $\mathbf{v}$,

$$\mathbf{v}^E = [v_0^E, v_1^E, v_2^E, \dots, v_{M-1}^E]^T = [v_0, v_2, v_4, \dots, v_{2M-2}]^T$$

and

$$\mathbf{v}^O = [v_0^O, v_1^O, v_2^O, \dots, v_{M-1}^O]^T = [v_1, v_3, v_5, \dots, v_{2M-1}]^T.$$

- Then compute the M th order alternative DFTs

$$[V_0^E, V_1^E, V_2^E, \dots, V_{M-1}^E]^T = \mathbf{V}^E = \mathcal{D}_N \mathbf{v}^E$$

and

$$[V_0^O, V_1^O, V_2^O, \dots, V_{M-1}^O]^T = \mathbf{V}^O = \mathcal{D}_N \mathbf{v}^O.$$

- Then construct the N th order sequence $\mathbf{V} = [V_0, V_1, V_2, \dots, V_{N-1}]^T$ from $\mathbf{V}^E$ and $\mathbf{V}^O$ via the **butterfly relations**

$$V_n = \begin{cases} V_n^E + w^n V_n^O & \text{for } n = 0, 1, 2, \dots, M-1 \\ V_{n-M}^E - w^{n-M} V_{n-M}^O & \text{for } n = M, M+1, M+2, \dots, N-1 \end{cases}$$

where $w = e^{-i2\pi/N}$.

2. (*Full radix 2 FFT*) This requires $N = 2^P$ for some $P \in \mathbb{N}$. To simplify the description, let

$$\sigma_k \in \begin{cases} \{0\} & \text{if } k = 0 \\ \{E, O\} & \text{if } k = 1, 2, 3, \dots, P \end{cases}.$$

- Recursively, split $\mathbf{v}$ into the sequences in the set

$$\{\mathbf{v}^{\sigma_0 \sigma_1 \sigma_2 \dots \sigma_K} : K = 0, 1, 2, \dots, P\}$$

where $\mathbf{v}^{\sigma_0} = \mathbf{v}$ and, letting $\Sigma = \sigma_0 \sigma_1 \sigma_2 \dots \sigma_K$ and $M = 2^{P-K-1}$,

$$\mathbf{v}^{\Sigma E} = [v_0^{\Sigma E}, v_1^{\Sigma E}, v_2^{\Sigma E}, \dots, v_{M-1}^{\Sigma E}]^T = [v_0^{\Sigma}, v_2^{\Sigma}, v_4^{\Sigma}, \dots, v_{2M-2}^{\Sigma}]^T$$

and

$$\mathbf{v}^{\Sigma O} = [v_0^{\Sigma O}, v_1^{\Sigma O}, v_2^{\Sigma O}, \dots, v_{M-1}^{\Sigma O}]^T = [v_1^{\Sigma}, v_3^{\Sigma}, v_5^{\Sigma}, \dots, v_{2M-1}^{\Sigma}]^T.$$

- For each of the 2^P first order sequences in $\{\mathbf{v}^{\sigma_0 \sigma_1 \sigma_2 \dots \sigma_P} = [v^{\sigma_1 \sigma_2 \dots \sigma_P}]\}$, set

$$\mathbf{V}^{\sigma_1 \sigma_2 \dots \sigma_P} = [V^{\sigma_1 \sigma_2 \dots \sigma_P}] = [v^{\sigma_1 \sigma_2 \dots \sigma_P}]$$

- Set $\mathbf{V} = \mathbf{V}^{\sigma_0}$ where

$$\{\mathbf{V}^{\sigma_0 \sigma_1 \sigma_2 \dots \sigma_K} : K = 0, 1, 2, \dots, P\}$$

is the set of sequences recursively constructed via the **butterfly relations**

$$V_n^{\Sigma} = \begin{cases} V_n^{\Sigma E} + w^n V_n^{\Sigma O} & \text{for } n = 0, 1, 2, \dots, M-1 \\ V_{n-M}^{\Sigma E} - w^{n-M} V_{n-M}^{\Sigma O} & \text{for } n = M, M+1, M+2, \dots, 2M-1 \end{cases}$$

where $\Sigma = \sigma_0 \sigma_1 \sigma_2 \dots \sigma_K$, $M = 2^{P-K-1}$, and $w = e^{(-i2\pi/N)2^K}$.

Facts:

All the following facts except those with specific reference can be found in [BH95], [CG00], or [Wal96].

1. Let $\mathbf{V}$ be the sequence computed from an N th order sequence $\mathbf{v}$ by either of the two algorithms above. Then $\mathbf{V} = \mathcal{D}_N \mathbf{v}$. Multiplying $\mathbf{V}$ by $N^{-1/2}$ yields $\widehat{\mathcal{F}}_N \mathbf{v}$.
2. For corresponding “fast” algorithms for computing the inverse DFT of $\mathbf{v}$, replace $w = e^{-i2\pi/N}$ and $w = e^{(-i2\pi/N)2^K}$, respectively, with $w = e^{i2\pi/N}$ and $w = e^{(i2\pi/N)2^K}$ in the above algorithms.
3. [BH95, p. 393] Assume $N = 2^P \gg 1$ for some $P \in \mathbb{N}$. Using just the defining formulas, the computation of an N th order DFT requires approximately $2N^2$ arithmetic operations ($(N-1)^2$ multiplications and $N(N-1)$ additions). Using the above described radix 2 FFT allows this same DFT to be computed with approximately $(3N/2) \log_2 N$ arithmetic operations ($N \log_2 N$ additions and $(N/2) \log_2 N$ multiplications, plus another N multiplications by $N^{-1/2}$ if it is desired to convert

the alternative DFT to the one previously described). Thus, the ratio of the number of arithmetic operations used to compute an N th order DFT by the two methods is

$$\frac{\text{operations if using the FFT}}{\text{operations if not using the FFT}} \approx \frac{3 \log_2 N}{4N}.$$

4. Assume $N = 2^P \gg 1$ for some $P \in \mathbb{N}$. Simply using the basic formulas, the computation of the discrete convolution of two sequences of order N ,

$$[\mathbf{v} * \mathbf{w}]_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} v_{k-j} w_j,$$

requires approximately $2N^2$ arithmetic operations ($(N+1)N$ multiplies and $(N-1)N$ additions). On the other hand, this convolution can be computed using the convolution identity

$$\mathbf{v} * \mathbf{w} = \widehat{\mathcal{F}}_N^{-1}[\mathbf{V}\mathbf{W}] \quad \text{where} \quad \mathbf{V} = \widehat{\mathcal{F}}_N \mathbf{v} \quad \text{and} \quad \mathbf{W} = \widehat{\mathcal{F}}_N \mathbf{w}.$$

Employing this identity along with the radix 2 FFT to compute the three DFTs involved requires approximately

- $3 \times (3N/2) \log_2 N$ arithmetic operations for the DFTs,
- Plus N multiplications for the product $\mathbf{V}\mathbf{W}$,

for a total of approximately $(9N/2) \log_2 N$ arithmetic operations. The ratio of operations required to compute this convolution by the two methods is

$$\frac{\text{operations if using identities and the FFT}}{\text{operations if not using identities and the FFT}} \approx \frac{9 \log_2 N}{4N}.$$

Examples:

1. Consider finding the DFT of

$$\mathbf{v} = [v_0, v_1, v_2, v_3]^T = [1, 2, 3, 4]^T$$

using the full radix 2 FFT. The order here is $N = 4$ (so $P = 2$). Recursively splitting $\mathbf{v}$ into the even and odd index subsequences yields

$$\mathbf{v}^E = [1, 3]^T \quad \text{and} \quad \mathbf{v}^O = [2, 4]^T$$

and then

$$\mathbf{v}^{EE} = [1], \quad \mathbf{v}^{EO} = [3], \quad \mathbf{v}^{OE} = [2], \quad \text{and} \quad \mathbf{v}^{OO} = [4].$$

(The $\sigma_0 = 0$ superscript is being suppressed.) Setting $\mathbf{V}^{\sigma_1 \sigma_2} = \mathbf{v}^{\sigma_1 \sigma_2}$ yields

$$\mathbf{V}^{EE} = [1], \quad \mathbf{V}^{EO} = [3], \quad \mathbf{V}^{OE} = [2], \quad \text{and} \quad \mathbf{V}^{OO} = [4].$$

Applying the first round of butterfly relations (in which $K = 1$, $M = 2^{P-K-1} = 1$ and $w = e^{(-i2\pi/4)2^1} = -1$),

$$\begin{aligned} V_0^E &= V_0^{EE} + w^0 V_0^{EO} = 1 + (-1)^0 3 = 4 \\ V_1^E &= V_0^{EE} - w^{1-1} V_0^{EO} = 1 - (-1)^0 3 = -2 \\ V_0^O &= V_0^{OE} + w^0 V_0^{OO} = 2 + (-1)^0 4 = 6 \\ V_1^O &= V_0^{OE} - w^{1-1} V_0^{OO} = 2 - (-1)^0 4 = -2, \end{aligned}$$

yields the second order sequences

$$\mathbf{V}^E = [4, -2]^T \quad \text{and} \quad \mathbf{V}^O = [6, -2]^T.$$

Applying the second (and last) round of butterfly relations (in which $K = 0$, $M = 2^{P-K-1} = 2$ and $w = e^{(-i2\pi/4)2^0} = -i$),

$$\begin{aligned} V_0 &= V_0^E + w^0 V_0^O = 4 + (-i)^0 6 = 10 \\ V_1 &= V_1^E + w^1 V_1^O = -2 + (-i)^1 (-2) = -2 + 2i \\ V_2 &= V_{2-2}^E - w^{2-2} V_{2-2}^O = 4 - (-i)^0 6 = -2 \\ V_3 &= V_{3-2}^E - w^{3-2} V_{3-2}^O = -2 - (-i)^1 (-2) = -2 - 2i, \end{aligned}$$

yields the alternative DFT

$$\mathbf{V} = \mathcal{D}\mathbf{v} = [10, -2 + 2i, -2, -2 - 2i]^T.$$

2. Consider the computation of an N th order DFT. When $N = 1024 = 2^{10}$, the ratio of operations required to compute this DFT with and without using the FFT is

$$\frac{\text{operations if using the FFT}}{\text{operations if not using the FFT}} \approx \frac{3 \cdot 10}{4 \cdot 2^{10}} \approx 7.3 \times 10^{-3}.$$

When $N = 11,048,576 = 2^{20}$, this ratio is

$$\frac{\text{operations if using the FFT}}{\text{operations if not using the FFT}} \approx \frac{3 \cdot 20}{4 \cdot 2^{20}} \approx 1.4 \times 10^{-5}.$$

3. Consider computing the convolution of two N th order sequences. When $N = 1024 = 2^{10}$, the ratio of operations required to compute this convolution with and without using the FFT and identities is

$$\frac{\text{operations if using identities and the FFT}}{\text{operations if not using identities and the FFT}} \approx \frac{9 \cdot 10}{4 \cdot 2^{10}} \approx 2.2 \times 10^{-2}.$$

When $N = 11,048,576 = 2^{20}$, this ratio is

$$\frac{\text{operations if using identities and the FFT}}{\text{operations if not using identities and the FFT}} \approx \frac{9 \cdot 20}{4 \cdot 2^{20}} \approx 4.3 \times 10^{-5}.$$

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Linear Algebra and Mathematical Physics

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59.1 Introduction

Linear algebra appears throughout physics. Linear differential equations, both ordinary and partial, appear through classical and quantum physics. Even where the equations are nonlinear, linear approximations are extremely powerful.

Two big ideas underpin linear analysis in physics. The first is the *Superposition Principle*. Suppose we have a linear problem where we need to compute the output for an arbitrary input. If there is a solution to the problem with input I_1 and output O_1 , a solution with input I_2 and output O_2 , etc., then the response to the input $c_1 I_1 + \cdots + c_k I_k$ is $c_1 O_1 + \cdots + c_k O_k$. It is, therefore, enough to solve our problem for a limited set of inputs I_k , as long as an arbitrary input can be written as a linear combination of these special cases.

The second big idea is the *Decoupling Principle*. If a system of coupled differential equations (or difference equations) involves a diagonalizable square matrix A , then it is helpful to pick new coordinates $\mathbf{y} = [\mathbf{x}]_{\mathcal{B}}$, where $\mathcal{B}$ is a basis of eigenvectors of A . Rewriting our equations in terms of the y variables, we discover that the evolution of each variable y_k depends only on y_k , and not on the other variables, and that the form of the equation depends only on the k th eigenvalue of A . We can then solve our equations, one variable at a time, to get $\mathbf{y}$ as a function of time and, hence, get $\mathbf{x}$ as a function of time. (When A is not diagonalizable, one uses a basis for which $[A]_{\mathcal{B}}$ is in Jordan canonical form. The resulting equations for $\mathbf{y}$ are not completely decoupled, but are still relatively easy.)

Thanks to Newton's Law, $F = ma$, much of classical physics is expressed in terms of systems of second order ordinary differential equations. If the force is a linear function of position, the resulting equations are linear, and the special solutions that come from eigenvectors of the force matrix are called *normal modes of oscillation*. For nonlinear problems near equilibrium, the force can always be expanded in a Taylor series, and for small oscillations the leading (linear) term is dominant. Solutions to realistic nonlinear problems, such as small oscillations of a pendulum, are then closely approximated by solutions to linear problems.

Linear field equations also permeate classical physics. Maxwell's equations, which govern electromagnetism, are linear. There are an infinite number of degrees of freedom, namely the value of the field at each point, but the Superposition Principle and the Decoupling Principle still apply. We use a continuous basis of possible inputs, namely Dirac δ functions, and the resulting outputs are called Green's functions. The response to an arbitrary input is then the convolution of the input and the relevant Green's function.

Nonrelativistic quantum mechanics is governed by Schrödinger's equation, which is also linear. Much of quantum mechanics reduces to diagonalizing the Hamiltonian operator and applying the Decoupling Principle.

Symmetry plays a big role in quantum mechanics. Both vectors and operators decompose into representations of the rotation groups $SO(3)$ and $SU(2)$. The irreducible representations are finite-dimensional, so the study of rotations (and angular momentum) often reduces to a study of finite matrices.

59.2 Normal Modes of Oscillation

Suppose we have two blocks, each with mass m , attached to three springs, as in Figure 59.1, with the spring constants as shown, and let $x_i(t)$ be the displacement of the i th block from equilibrium at time t . It is easy to see that if $x_1(0) = x_2(0)$, and if $\dot{x}_1(0) = \dot{x}_2(0)$, then $x_1(t) = x_2(t)$ for all time. The middle spring never gets stretched, and the two blocks oscillate, in phase, with angular frequency $\omega_1 = \sqrt{k_1/m}$. If $x_2(0) = -x_1(0)$ and $\dot{x}_2(0) = -\dot{x}_1(0)$, then by symmetry $x_2(t) = -x_1(t)$ for all time, and each block oscillates with angular frequency $\omega_2 = \sqrt{(k_1 + 2k_2)/m}$. (This example is worked out in detail below.) These two solutions, with $x_1(t) = \pm x_2(t)$, are called *normal modes of oscillation*. Remarkably, every solution to the equations of motion is a linear combination of these two normal modes.

Definitions:

Suppose we have an arrangement of blocks, all of the same mass m , and springs with varying spring constants. Let $x_1(t), \dots, x_n(t)$ denote the locations of the blocks, relative to equilibrium, and $\mathbf{x} = [x_1, \dots, x_n]^T$. For any function $f(t)$, let $\dot{f}(t) = df/dt$. The **kinetic energy** is $T = m \sum_k \dot{x}_k^2/2$. The **potential energy** is

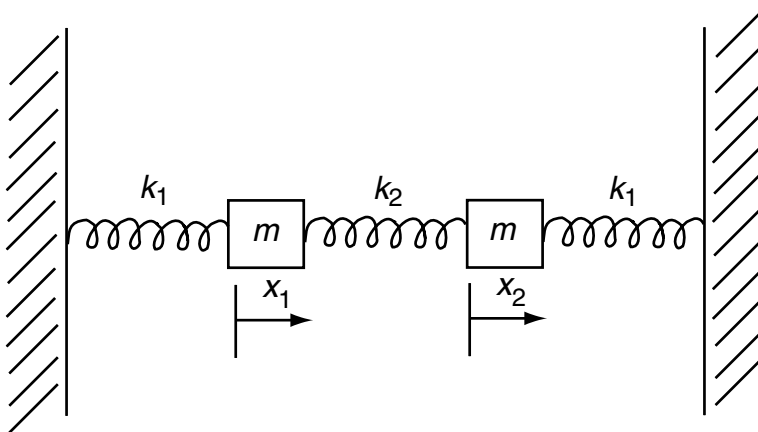


FIGURE 59.1 Coupled oscillators.

$V(\mathbf{x}) = \sum_{ij} a_{ij} x_i x_j / 2$, where $A = (a_{ij})$ is a symmetric matrix. The **equations of motion** are

$$m \frac{d^2 \mathbf{x}}{dt^2} = -A\mathbf{x}.$$

Let $\mathcal{B} = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ be a basis of eigenvectors of A , and let $\mathbf{y}(t) = [\mathbf{x}(t)]_{\mathcal{B}}$ be the coordinates of $\mathbf{x}(t)$ in this basis.

Facts:

(See Chapter 13 of [Mar70], Chapter 6 of [Gol80], and Chapter 5 of [Sad01].)

1. A is diagonalizable, and the eigenvalues of A are all real.
2. The eigenvectors can be chosen orthonormal with respect to the standard inner product: $\langle \mathbf{z}_i, \mathbf{z}_j \rangle = \delta_{ij}$.
3. The initial conditions for $\mathbf{y}(t)$ can be computed using the inner product: $y_k(0) = \langle \mathbf{z}_k, \mathbf{x}(0) \rangle$, $\dot{y}_k(0) = \langle \mathbf{z}_k, \dot{\mathbf{x}}(0) \rangle$.
4. In terms of the $\mathbf{y}$ variables, the equations of motion reduce to $m d^2 y_k / dt^2 = -\lambda_k y_k$, where λ_k is the eigenvalue corresponding to the eigenvector $\mathbf{z}_k$.
5. The solution to this equation depends on the sign of λ_k . If $\lambda_k > 0$, set $\omega_k = \sqrt{\lambda_k / m}$. We then have

$$y_k(t) = y_k(0) \cos(\omega_k t) + \frac{\dot{y}_k(0)}{\omega_k} \sin(\omega_k t).$$

If $\lambda_k < 0$, set $\kappa_k = \sqrt{-\lambda_k / m}$ and we have

$$y_k(t) = y_k(0) \cosh(\kappa_k t) + \frac{\dot{y}_k(0)}{\kappa_k} \sinh(\kappa_k t).$$

Finally, if $\lambda_k = 0$, then

$$y_k(t) = y_k(0) + \dot{y}_k(0)t.$$

6. If the system has translational symmetry, then there is a $\lambda = 0$ mode describing uniform motion of the system.
7. If the system has rotational symmetry, then there is a $\lambda = 0$ mode describing uniform rotation.
8. All solutions of the equations of motion are of the form $\mathbf{x}(t) = \sum y_k(t) \mathbf{z}_k$, where for each nonzero λ_k , $y_k(t)$ is of the form given in Fact 5.

Examples:

1. In the block-and-spring example above, the kinetic energy is $m(\dot{x}_1^2 + \dot{x}_2^2)/2$, while the potential energy is $(k_1 x_1^2 + k_2 (x_1 - x_2)^2 + k_1 x_2^2)/2 = \langle \mathbf{x}, A\mathbf{x} \rangle / 2$, where $A = \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_1 + k_2 \end{pmatrix}$. The eigenvalues of A are k_1 and $k_1 + 2k_2$, with normalized eigenvectors $(\sqrt{2}/2, \sqrt{2}/2)^T$ and $(\sqrt{2}/2, -\sqrt{2}/2)^T$. Both eigenvalues are positive, so we have oscillations with angular frequencies $\omega_1 = \sqrt{k_1/m}$ and $\omega_2 = \sqrt{(k_1 + 2k_2)/m}$. Suppose we start by pushing the first block to the right and letting go. That is, suppose $\mathbf{x}(0) = (1, 0)^T$ and $\dot{\mathbf{x}}(0) = (0, 0)^T$. From the initial data we compute

$$\begin{aligned}
y_1(0) &= \frac{\sqrt{2}}{2}(x_1(0) + x_2(0)) = \sqrt{2}/2 \\
y_2(0) &= \frac{\sqrt{2}}{2}(x_1(0) - x_2(0)) = \sqrt{2}/2 \\
\dot{y}_1(0) &= \frac{\sqrt{2}}{2}(\dot{x}_1(0) + \dot{x}_2(0)) = 0 \\
\dot{y}_2(0) &= \frac{\sqrt{2}}{2}(\dot{x}_1(0) - \dot{x}_2(0)) = 0 \\
y_1(t) &= y_1(0) \cos(\omega_1 t) + \frac{\dot{y}_1(0)}{\omega_1} \sin(\omega_1 t) = \sqrt{2} \cos(\omega_1 t)/2 \\
y_2(t) &= y_2(0) \cos(\omega_2 t) + \frac{\dot{y}_2(0)}{\omega_2} \sin(\omega_2 t) = \sqrt{2} \cos(\omega_2 t)/2 \\
\mathbf{x}(t) &= y_1(t)\mathbf{z}_1 + y_2(t)\mathbf{z}_2 = \frac{1}{2} \begin{pmatrix} \cos(\omega_1 t) + \cos(\omega_2 t) \\ \cos(\omega_1 t) - \cos(\omega_2 t) \end{pmatrix}.
\end{aligned}$$

2. LC circuits obey the same equations as blocks and springs, with the capacitances playing the role of spring constants and the inductances playing the role of mass, and with the current around each loop playing the role of x_i .
3. Small oscillations: A particle in an arbitrary potential $V(\mathbf{x})$, or a system of identical-mass particles in an arbitrary n -body potential, follows the equation $m d^2 \mathbf{x} / dt^2 = -\nabla V(\mathbf{x})$. If $\mathbf{x} = \mathbf{x}_0$ is a critical point of the potential, so $\nabla V(\mathbf{x}_0) = 0$, then we expand $V(\mathbf{x})$ around $\mathbf{x} = \mathbf{x}_0$ in a Taylor series:

$$V(\mathbf{x}) = V(\mathbf{x}_0) + \frac{1}{2} \sum_{ij} a_{ij} (\mathbf{x} - \mathbf{x}_0)_i (\mathbf{x} - \mathbf{x}_0)_j + O(|\mathbf{x} - \mathbf{x}_0|^3),$$

where $a_{ij} = \left. \frac{\partial^2 V}{\partial x_i \partial x_j} \right|_{\mathbf{x}=\mathbf{x}_0}$, so $\nabla V(\mathbf{x}) = A(\mathbf{x} - \mathbf{x}_0) + O(|\mathbf{x} - \mathbf{x}_0|^2)$, and our displacement $\mathbf{x} - \mathbf{x}_0$ from equilibrium is governed by the approximate equation $m d^2 (\mathbf{x} - \mathbf{x}_0) / dt^2 = -A(\mathbf{x} - \mathbf{x}_0)$.

For example, a pendulum of mass m and length ℓ has quadratic kinetic energy $m\ell^2 \dot{\theta}^2 / 2$ and nonlinear potential energy $mg\ell(1 - \cos(\theta))$. For θ small, this potential energy is approximated by $mg\ell\theta^2/2$, and the equations of motion are approximated by $\ell d^2 \theta / dt^2 = -g\theta$, and yields oscillations of angular frequency $\sqrt{g/\ell}$. The same ideas apply to motion of a pendulum near the top of the circle: $\theta = \pi$. Then $V(\theta) \approx mg\ell(2 - (\theta - \pi)^2/2)$, and our equations of motion are approximately $\ell d^2 (\theta - \pi) / dt^2 = +g(\theta - \pi)$. The deviation of θ from the unstable equilibrium grows as $e^{\kappa t}$, with $\kappa = \sqrt{g/\ell}$, until $\theta - \pi$ is large enough that our quadratic approximation for $V(\theta)$ is no longer valid.

Finally, one can consider two pendula, near their stable equilibria, attached by a weak spring. The resulting equations are almost identical to those of the coupled springs of Figure 59.1.

4. Central force motion (see Chapter 3 of [Gol80] or Chapter 8 of [Mar70]): In systems with symmetry, it is often possible to use conserved quantities to integrate out some of the variables, obtaining reduced equations for the remaining variables. For instance, if an object is moving in a central force (e.g., a planet around a star or a classical electron around the nucleus), conservation of angular momentum allows us to integrate out the angular variables and get an equation for the distance r . The radius then oscillates in a pseudopotential $V(r)$, obtained by adding a $1/r^2$ centrifugal term to the true potential $V_0(r)$. Orbits that are almost circular are described by small oscillations of the variable r around the minimum of the pseudopotential $V(r)$, and the frequency of oscillation is $\sqrt{V''(r_0)/m}$, where the pseudopotential has a minimum at $r = r_0$. When the true potential is a $1/r$ attraction (as with gravitation and electromagnetism), these oscillations have the same period as the orbital motion itself. Planets traverse elliptical orbits, with the sun at a focus, and the nearest approach to the sun (the perihelion) occurs at the same point each year. When the true potential is an r^2 attraction (simple harmonic motion), the radial oscillations occur with frequency twice that of the orbit. The motion is elliptical with the center of force at the center of the orbit, and there are

two perihelia per cycle. For almost any other kind of force, the radial oscillations and the rotation are incommensurate, the orbit is not a closed curve, and the perihelion precesses.

59.3 Lagrangian Mechanics

In the previous section, we assumed that all the particles had the same mass or, equivalently, that the kinetic energy was proportional to the squared norm of the velocity. Here we relax this assumption and we also allow generalized coordinates.

Definitions:

The **Lagrangian function** is $L(\mathbf{q}, \dot{\mathbf{q}}) = T - V$, where T is the kinetic energy and V is the potential energy. One can express the Lagrangian in terms of arbitrary generalized coordinates $\mathbf{q}$ and their derivatives $\dot{\mathbf{q}}$. The **kinetic energy** is typically quadratic in the velocity: $T = \langle \dot{\mathbf{q}}, B(\mathbf{q})\dot{\mathbf{q}} \rangle / 2$, where the symmetric “mass matrix” B may depend on the coordinates $\mathbf{q}$, but not on the velocities $\dot{\mathbf{q}}$. The **potential energy** V depends only on the coordinates $\mathbf{q}$ (and not on $\dot{\mathbf{q}}$), but may be nonlinear.

If $\mathbf{q}_0$ is a critical point of V , we consider motion with $\mathbf{q}$ close to $\mathbf{q}_0$ and $\dot{\mathbf{q}}$ small.

Facts:

(See Chapter 7 of [Mar70] or Chapters 2 and 6 of [Gol80].)

1. The Euler–Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) = \frac{\partial L}{\partial q_k}$$

reduce to the approximate equations of motion

$$B \frac{d^2(\mathbf{q} - \mathbf{q}_0)}{dt^2} = -A(\mathbf{q} - \mathbf{q}_0),$$

where $a_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\mathbf{q}=\mathbf{q}_0}$, essentially as before, and the mass matrix B is evaluated at $\mathbf{q} = \mathbf{q}_0$. Instead of looking for eigenvalues and eigenvectors of A , we look for numbers λ_k and vectors $\mathbf{z}_k$ such that $A\mathbf{z}_k = \lambda_k B\mathbf{z}_k$. (See Chapter 43.) We then let $\mathbf{y} = [\mathbf{q} - \mathbf{q}_0]_B$.

2. The matrices A and B are symmetric, and the eigenvalues of B are all positive.
3. The numbers λ_k are the roots of the polynomial $\det(xB - A)$. When B is the identity matrix, these reduce to the eigenvalues of A .
4. One can find a basis of solutions $\mathbf{z}_k$ to $A\mathbf{z}_k = \lambda_k B\mathbf{z}_k$, with λ_k real. The numbers λ_k are the eigenvalues of $B^{-1}A$, or equivalently of the symmetric matrix $B^{-1/2}AB^{-1/2}$, which explains why the λ_k 's are real.
5. The eigenvectors can be chosen orthonormal with respect to an inner product involving B . (See Chapter 5.) That is, if $\langle \mathbf{u}, \mathbf{v} \rangle_B = \mathbf{u}^T B \mathbf{v}$, then $\langle \mathbf{z}_i, \mathbf{z}_j \rangle_B = \delta_{ij}$.
6. The initial conditions for $\mathbf{y}(t)$ can be computed using the modified inner product of the previous fact: $y_k(0) = \langle \mathbf{z}_k, \mathbf{q}(0) - \mathbf{q}_0(0) \rangle_B$, $\dot{y}_k(0) = \langle \mathbf{z}_k, \dot{\mathbf{q}}(0) \rangle_B$.
7. In terms of the $\mathbf{y}$ variables, the approximate equations of motion reduce to the decoupled equations $d^2 y_k / dt^2 = -\lambda_k y_k$.
8. The solution to these equations depends on the sign of λ_k . If $\lambda_k > 0$, set $\omega_k = \sqrt{\lambda_k}$; if $\lambda_k < 0$, set $\kappa_k = \sqrt{-\lambda_k}$. With values of ω_k or κ_k , these the solutions take the same form as in Fact 5 of section 59.2.
9. If the system is symmetric under the action of a continuous group, then there is a $\lambda = 0$ mode for each generator of this group.

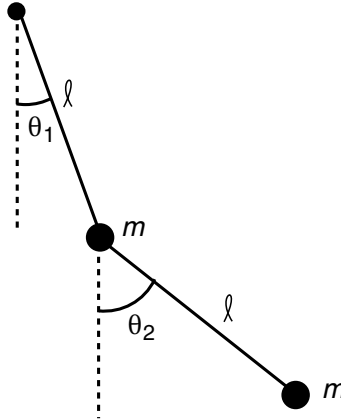


FIGURE 59.2 A double pendulum.

10. All solutions of the approximate equations of motion are of the form $\mathbf{q}(t) = \mathbf{q}_0 + \sum y_k(t)\mathbf{z}_k$, where for each nonzero λ_k , $y_k(t)$ is of the form given by Fact 8 above (and Fact 5 of Section 59.2).

Examples:

- Consider the double pendulum of Figure 59.2, where each ball has mass m and each rod has length ℓ . For large motions, this system is famously chaotic, but for small oscillations it is simple. The two coordinates are the angles θ_1 and θ_2 , and the potential energy of the system is $mg\ell(3 - 2\cos(\theta_1) - \cos(\theta_2)) \approx mg\ell(\theta_1^2 + \theta_2^2/2)$, so $A = mg\ell \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. The kinetic energy is $\frac{m\ell^2}{2}(\dot{\theta}_1^2 + (\sin(\theta_1)\dot{\theta}_1 + \sin(\theta_2)\dot{\theta}_2)^2 + (\cos(\theta_1)\dot{\theta}_1 + \cos(\theta_2)\dot{\theta}_2)^2)$. For small values of θ_1 and θ_2 , this is approximately $\frac{m\ell^2}{2}(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2)$, so $B = m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. $\det(xB - A) = m^2\ell^4(x^2 - 4(g/\ell)x + 2g^2/\ell^2)$, with roots $\lambda_1 = (g/\ell)(2 + \sqrt{2})$ and $\lambda_2 = (g/\ell)(2 - \sqrt{2})$, and with $\mathbf{z}_i = c_i(1, \mp\sqrt{2})^T$, $i = 1, 2$, where c_i are normalization constants. The two normal modes are as follows: There is a fast mode, with $\omega_1 = \sqrt{(g/\ell)(2 + \sqrt{2})} \approx 1.8478\sqrt{g/\ell}$, with the two pendula swinging in opposite directions, and with the bottom pendulum swinging $\sqrt{2}$ more than the top; there is a slow mode, with $\omega_2 \sqrt{(g/\ell)(2 - \sqrt{2})} \approx 0.7654\sqrt{g/\ell}$, with the two pendula swinging in the same direction, and with the bottom pendulum swinging $\sqrt{2}$ more than the top.

59.4 Schrödinger's Equation

In quantum mechanics, the evolution of a particle of mass m , moving in a time-dependent potential $V(x, t)$, is described by Schrödinger's equation,

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(x, t) + V(x, t) \psi(x, t),$$

where $\hbar$ is Planck's constant divided by 2π , and the square of the complex wavefunction $\psi(x, t)$ describes the probability of finding a particle at position x at time t . Space and time are not treated on equal footing. We consider the wavefunction ψ to be a square-integrable function of x that evolves in time.

Definitions:

Let $\mathcal{H} = L^2(\mathbb{R}^n)$ be the Hilbert space of square-integrable functions on $\mathbb{R}^n$ with “inner product”

$$\langle \phi | \psi \rangle = \int_{\mathbb{R}^n} \overline{\phi(x)} \psi(x) d^n x.$$

Note that this inner product is linear in the *second* factor and conjugate-linear in the first factor. Although mathematicians usually choose their complex inner products $\langle u, v \rangle$ to be linear in u and conjugate-linear in v , among physicists the convention, and notation, is invariably that of the above equation. The bracket of ϕ and ψ can be viewed as a pairing of two pieces, the “bra” $\langle \phi |$ and the “ket” $|\psi\rangle$. The ket $|\psi\rangle$ is a vector in $\mathcal{H}$, while $\langle \phi |$ is a map from $\mathcal{H}$ to the complex numbers, namely, “take the inner product of ϕ with an input vector.”

The Hermitian adjoint of an operator A , denoted A^* , is the unique operator such that $\langle A^* \phi | \psi \rangle = \langle \phi | A \psi \rangle$, for all vectors ϕ, ψ . An operator A is called **Hermitian**, or **self-adjoint**, if $A^* = A$, and **unitary** if $A^* = A^{-1}$.

The **commutator** of two operators A and B , denoted $[A, B]$, is the difference $AB - BA$. A and B are said to **commute** if $AB = BA$.

The **expectation value** of an operator A in the state $|\psi\rangle$ is the statistical average of repeated measurements of A in the state $|\psi\rangle$, and is denoted $\langle A \rangle$, with the dependence on $|\psi\rangle$ implicit. The **uncertainty in A** , denoted ΔA , is the root mean squared variation in measurements of A .

The **generalized eigenvalues** of an operator A are points in the spectrum of A , and the **generalized eigenvectors** are formal solutions to $A|\psi\rangle = \lambda|\psi\rangle$. These may not be true eigenvalues and eigenvectors if ψ is not square integrable. (See Facts 11 to 15 below.) This use of the term “generalized eigenvector,” which is standard in physics, has *nothing* to do with the same term in matrix theory (where it signifies vectors $\mathbf{v}$ for which $(A - \lambda I)^k \mathbf{v} = 0$ for some positive integer k).

Facts:

(See Chapter 6 of [Sch68] or Chapters 5 and 7 of [Mes00].)

1. The Schrödinger equation can be recast as an ordinary differential equation with values in $\mathcal{H}$:

$$i\hbar \frac{d|\psi\rangle}{dt} = H(t)|\psi\rangle,$$

where $H = -\frac{\hbar^2}{2m} \nabla^2 + V$ is the Hamiltonian operator.

2. Physically measurable quantities, also called *observables*, are represented by Hermitian operators. It is easy to see that the position operator $(X\psi)(x) = x\psi(x)$, the momentum operator $(P\psi)(x) = -i\hbar \nabla \psi(x)$, and the Hamiltonian $H = P^2/2m + V$ are all self-adjoint.
3. If an observable a is represented by the operator A , then the possible values of a measurement of a are the (generalized) eigenvalues of A .
4. Two Hermitian operators A, B can be simultaneously diagonalized if and only if they commute.
5. Suppose the state of the system is described by the vector $|\psi\rangle = \sum_n c_n |\phi_n\rangle$, where $\sum |c_n|^2 = 1$ and each $|\phi_n\rangle$ is a normalized eigenvector of A with eigenvalue λ_n . Then the probability of a measurement of a yielding the value λ_n is $|c_n|^2$.
6. If $|\psi\rangle$ is as in the previous Fact, then the expectation value of A is $\langle A \rangle = \sum_n \lambda_n |c_n|^2$.
7. The uncertainty of A satisfies $(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2$.
8. If A, B , and C are Hermitian operators with $[A, B] = iC$, then $\Delta A \Delta B \geq |\langle C \rangle|/2$.
9. In particular, $XP - PX = i\hbar$, so $\Delta X \Delta P \geq \hbar/2$. This is *Heisenberg’s uncertainty principle*.
10. If the Hamiltonian operator does not depend on time, then energy is conserved. In fact, if $|\psi(0)\rangle = \sum c_n |\phi_n\rangle$, where $H|\phi_n\rangle = E_n |\phi_n\rangle$, then $|\psi(t)\rangle = \sum c_n e^{-iE_n t/\hbar} |\phi_n\rangle$. (Eigenvalues of the Hamiltonian are usually denoted E_n , for energy.) Solving the Schrödinger equation is tantamount to diagonalizing the Hamiltonian and using a basis of eigenvectors.

11. Operators may have continuous spectrum, in which case the generalized eigenvectors are not square-integrable. In particular, e^{ikx} is a generalized eigenvector for $P = -i\hbar d/dx$ with generalized eigenvalue $\hbar k$, and the Dirac delta function $\delta(x - a)$ is a generalized eigenvector for X with eigenvalue a .
12. Let $|A, \alpha\rangle$ be a generalized eigenvector of the operator A with generalized eigenvalue α . If A has continuous spectrum, then the decomposition of a state $|\psi\rangle$ involves integrating over eigenvalues instead of summing: $|\psi\rangle = \int f(\alpha)|A, \alpha\rangle d\alpha$. The generalized eigenstates are usually normalized so that $\langle\psi|\psi\rangle = \int |f(\alpha)|^2 d\alpha$. Equivalently, $\langle A, \alpha|A, \beta\rangle = \delta(\alpha - \beta)$.
13. For continuous spectra, $|f(\alpha)|^2$ is not the probability of a measurement of A yielding the value α . Rather, $|f(\alpha)|^2$ is a probability *density*, and the probability of a measurement yielding a value between a and b is $\int_a^b |f(\alpha)|^2 d\alpha$.
14. The two most common expansions in terms of generalized eigenvalues are for the position operator and the momentum operator: $|\psi\rangle = \int \psi(x)|X, x\rangle dx = \int \hat{\psi}(k)|P, \hbar k\rangle dk$. The coefficients $\hat{\psi}(k)$ of $|\psi\rangle$ in the momentum basis are the Fourier transform of the coefficients $\psi(x)$ in the position basis. From this perspective, the Fourier transform is just a change-of-basis.
15. An operator may have both discrete and continuous spectrum, in which case eigenfunction expansions involve summing over discrete eigenvalues and integrating over continuous eigenvalues. For example, for the Hamiltonian of a hydrogen atom, there are discrete negative eigenvalues that describe bound states, and a continuum of positive eigenvalues that describe ionized hydrogen, with the electron having broken free of the nucleus.

Examples:

1. *The one dimensional harmonic oscillator.* We have seen that a classical harmonic oscillator with potential energy $kx^2/2$ has frequency $\omega = \sqrt{k/m}$, so we write the Hamiltonian of a quantum mechanical harmonic oscillator as

$$H = \frac{P^2}{2m} + \frac{kX^2}{2} = \frac{P^2 + m^2\omega^2 X^2}{2m}.$$

We will compute the eigenvalues and eigenvectors of this Hamiltonian.

We define a *lowering operator*

$$a = \frac{P - im\omega X}{\sqrt{2m\hbar\omega}}.$$

The Hermitian conjugate of a is the *raising operator*

$$a^* = \frac{P + im\omega X}{\sqrt{2m\hbar\omega}}.$$

Note that a and a^* do not commute. Rather, $[a, a^*] = 1$. In terms of a and a^* , the Hamiltonian takes the form

$$H = \hbar\omega(a^*a + aa^*)/2 = \frac{\hbar\omega}{2}(2a^*a + 1) = \frac{\hbar\omega}{2}(2aa^* - 1).$$

Note that a^*a is positive-definite, since $\langle\psi|a^*a|\psi\rangle = \langle a\psi|a\psi\rangle \geq 0$, so the eigenvalues of energy must all be at least $\hbar\omega/2$.

Since $Ha = a(H - \hbar\omega)$, the operator a serves to lower the energy of a state by $\hbar\omega$. If $|\phi\rangle$ is an eigenvector of H with eigenvalue E , then $Ha|\phi\rangle = a(H - \hbar\omega)|\phi\rangle = a(E - \hbar\omega)|\phi\rangle = (E - \hbar\omega)a|\phi\rangle$, so $a|\phi\rangle$ is either the zero vector or a state with energy $E - \hbar\omega$. Since we cannot reduce the energy below $\hbar\omega/2$, by applying a repeatedly (say, $n \geq 0$ times), we must eventually get a vector $|\phi_0\rangle$ for which $a|\phi_0\rangle = 0$. But then $H|\phi_0\rangle = \frac{\hbar\omega}{2}(2a^*a + 1)|\phi_0\rangle = \frac{\hbar\omega}{2}|\phi_0\rangle$. Since it took n lowerings to get the energy to $\hbar\omega/2$, our original state $|\phi\rangle$ must have had energy $(n + \frac{1}{2})\hbar\omega$.

The notation $|n\rangle$ is often used for this n th excited state, so $a|n\rangle$ is a constant times $|n-1\rangle$. It remains to compute that constant. Normalizing $\langle n|n\rangle = 1$ and picking the phases such that $\langle n-1|an\rangle > 0$, we compute $\langle an|an\rangle = \langle n|a^*an\rangle = \frac{\langle n|Hn\rangle}{\hbar\omega} - \frac{1}{2} = n\langle n|n\rangle = n$, so $a|n\rangle = \sqrt{n}|n-1\rangle$. A similar calculation yields $a^*|n\rangle = \sqrt{n+1}|n+1\rangle$ and, hence, $|n\rangle = (a^*)^n|0\rangle/\sqrt{n!}$.

The state $|0\rangle$ is in the kernel of a . In a coordinate basis, where X is multiplication by x and $P = -i\hbar d/dx$, the equation $a|0\rangle = 0$ becomes a first-order differential equation

$$\frac{d\psi(x)}{dx} + \frac{m\omega x}{\hbar}\psi(x) = 0,$$

whose solution is the Gaussian $\psi(x) = \exp(-m\omega x^2/2\hbar)$ times a normalization constant. The n th state is obtained by applying the differential operator $\frac{d}{dx} - \frac{m\omega x}{\hbar}$ to the Gaussian n times. The result is an n th order polynomial in x times the Gaussian.

59.5 Angular Momentum and Representations of the Rotation Group

The same techniques that solved the harmonic oscillator also work to diagonalize the angular momentum operator.

Definitions:

Angular momentum is a vector: $\vec{L} = \vec{X} \times \vec{P}$, or in coordinates, $L_1 = X_2P_3 - X_3P_2$, $L_2 = X_3P_1 - X_1P_3$, $L_3 = X_1P_2 - X_2P_1$. Each L_i is a self-adjoint observable. We define $L^2 = L_1^2 + L_2^2 + L_3^2$, and define a **raising operator** $L_+ = L_1 + iL_2$ and a **lowering operator** $L_- = L_1 - iL_2$.

Facts:

(See Chapter 7 of [Sch68] or Chapter 13 of [Mes00].)

1. The three components of angular momentum do not commute. Rather,

$$[L_1, L_2] = i\hbar L_3, \quad [L_2, L_3] = i\hbar L_1, \quad [L_3, L_1] = i\hbar L_2.$$

By the uncertainty principle, this means that only one component of the angular momentum can be known at a time.

2. L^2 is Hermitian, and each $[L_i, L^2] = 0$. It is possible to know both L^2 and L_3 , and we consider simultaneous eigenstates $|\ell, m\rangle$ of L^2 and L_3 , where ℓ labels the eigenvalue of L^2 and $\hbar m$ is the eigenvalue of L_3 .
3. $[L^2, L_\pm] = 0$ and $[L_3, L_\pm] = \pm\hbar L_\pm$. This means that L_+ does not change the eigenvalue of L^2 , but increases the eigenvalue of L_3 by $\hbar$. Likewise, L_- decreases the eigenvalue of L_3 by $\hbar$.
4. Since $L^2 - L_3^2 = L_1^2 + L_2^2 \geq 0$, there is a limit to how big m (or $-m$) can get. For each ℓ , there is a state $|\ell, m_{\max}\rangle$ for which $L_+|\ell, m_{\max}\rangle = 0$, and a state $|\ell, m_{\min}\rangle$ for which $L_-|\ell, m_{\min}\rangle = 0$. We set the label ℓ to be equal to $m_{\max}$.
5. $L_-L_+ = L_1^2 + L_2^2 - \hbar L_3$ and $L_+L_- = L_1^2 + L_2^2 + \hbar L_3$, so we can write L^2 in terms of $L_\pm$ and L_3 : $L^2 = L_-L_+ + L_3^2 + \hbar L_3 = L_+L_- + L_3^2 - \hbar L_3$.
6. The minimum value of m is $-\ell$. Since $2\ell = m_{\max} - m_{\min}$ is an integer, ℓ must be half of a nonnegative integer.
7. The states $|\ell, m\rangle$, with m ranging from $-\ell$ to ℓ , form a $(2\ell + 1)$ dimensional irreducible representation of the Lie algebra of $SO(3)$. We denote this representation V_ℓ , and call it the “spin- ℓ ” representation.
8. In V_ℓ , we have $L^2|\ell, m\rangle = \hbar^2\ell(\ell + 1)|\ell, m\rangle$, $L_3|\ell, m\rangle = \hbar m|\ell, m\rangle$, and $L_\pm|\ell, m\rangle = \hbar\sqrt{\ell(\ell + 1) - m(m \pm 1)}|\ell, m \pm 1\rangle$.

9. If u is a unit vector, then a rotation by the angle θ about the u axis is implemented by the unitary operator $\exp(-i\theta L \cdot u/\hbar)$.
10. Since rotation by 2π equals the identity, representations of the Lie group $SO(3)$ satisfy the additional condition $\exp(2\pi i L_3) = 1$, which forces m (and, therefore, ℓ) to be an integer.
11. If one particle has angular momentum ℓ_1 and another has angular momentum ℓ_2 , then the combined angular momentum can be any integer between $|\ell_1 - \ell_2|$ and $\ell_1 + \ell_2$. In terms of representations, $V_{\ell_1} \otimes V_{\ell_2} = \bigoplus_{\ell=|\ell_1-\ell_2|}^{\ell_1+\ell_2} V_{\ell}$.
12. The Lie group $SU(2)$ is the double cover of $SO(3)$, and has the same Lie algebra. The generators are usually denoted J rather than L , and the maximum value of m is denoted j rather than ℓ , but otherwise the computations are the same. J describes the total angular momentum of a particle, including spin, and j can be either an integer or a half-integer.
13. Particles with j integral are called bosons, while particles with j half-integral are called fermions.
14. If the Hamiltonian is rotationally symmetric, then angular momentum is conserved, and our energy eigenstates can be chosen to be eigenstates of J^2 and J_3 .

59.6 Green's Functions

Expansions in a continuous basis of eigenfunctions are not limited to quantum mechanics. The Dirac δ is an eigenfunction of position, and any function can be written trivially as an integral over δ functions:

$$f(x) = \int f(y)\delta(x-y)dy = \int f(y)|X, y\rangle dy.$$

It, therefore, suffices to solve linear input–output problems in the case where the input is a δ -function located at an arbitrary point y . The resulting solution $G(y, x)$ is called a **Green's function** (or in some texts, **Green function**) for the problem, and the solution for an arbitrary input $f(x)$ is the convolution $\int f(y)G(y, x)dy$.

Facts:

(See [Jac75], especially Chapters 1 to 3, for many applications to electrostatics, and see Chapter 11 of [Sad01] for a general introduction to Green's functions.)

1. Green's functions are sometimes called *integral kernels*, especially in the mathematics literature, or *propagators* in quantum field theory. The term *propagator* is also sometimes used for the Fourier transform of a Green's function.
2. Linear partial differential equations appear throughout physics. Examples include Maxwell's equations, Laplace's equation, Schrödinger's equation, the heat equation, the wave equation, and the Dirac equation. Each equation generates its own Green's function.
3. Some boundary value problems involve Neumann boundary conditions, in which the normal derivative of a function (as opposed to the value of a function) is specified on S , and some problems involve mixed Neumann and Dirichlet conditions. The formalism for these cases is a simple modification of the Dirichlet formalism.
4. Two common techniques for computing Green's functions are Fourier transforms and the method of images.
5. Fourier transforms apply when the problem has translational symmetry, as in the heat equation example, above. We decompose a δ function as a linear combination of exponentials e^{ikx} , compute the response for each exponential, and re-sum.
6. The method of images is illustrated in Example 2, where the actual response $G(y, x)$ is a sum of two terms. The first is the response $G_0(y, x)$ to the actual charge at y , computed without boundary, and the second is the response to a mirror charge, located at a point outside D , and chosen so that the sum of the two terms is zero on S .

Examples:

1. *Electrostatics without boundaries.* The electrostatic potential $\phi(x)$ is governed by Poisson's equation:

$$\nabla^2 \phi = -4\pi \rho(x),$$

where $\rho(x)$ is the charge density. Here, ρ is the input and ϕ is the output. Since the solution to $\nabla^2 G(y, x) = -4\pi \delta^3(x - y)$ is $G(y, x) = |x - y|^{-1}$, the potential due to a charge distribution $\rho(x)$ is $\phi(x) = \int d^3 y \rho(y)/|x - y|$. (Note that, when we write $\nabla^2 G(y, x)$, we are taking the second derivative of $G(y, x)$ with respect to x . The variable y is just a parameter.)

2. *Electrostatics with boundary conditions.* Poisson's equation on a domain D with boundary S is more subtle, as we need to apply boundary conditions on S . Suppose that D is the exterior of a ball of radius R , and that we apply the homogeneous Dirichlet boundary condition $\phi = 0$ on S . (This corresponds to S being a grounded conducting sphere.) The function $G_0(y, x) = 1/|x - y|$ satisfies $\nabla^2 G_0(y, x) = -4\pi \delta^3(x - y)$, but does not satisfy the boundary condition. The function $G(y, x) = G_0(y, x) - \frac{R}{|y|} G_0(R^2 y/|y|^2, x)$ satisfies $\nabla^2 G(y, x) = -4\pi \delta^3(x - y)$ and $G(y, x) = 0$ for $x \in S$.

Nonzero boundary values can be considered part of the input. If we want to solve the equation $\nabla^2 \phi = -4\pi \rho$ on D with boundary values $f(x)$ on S , then we have two different Green's functions to compute. For each $y \in D$, we compute $G_1(y, x)$, the solution to $\nabla^2 G_1(y, x) = -4\pi \delta^3(x - y)$ with boundary value zero on S . For each $z \in S$, we compute $G_2(z, x)$, the solution to $\nabla^2 G_2(z, x) = 0$ on D with boundary value $\delta^2(x - z)$ on S . Our solution to the entire problem is then $\phi(x) = \int_D d^3 y G_1(y, x) \rho(y) + \int_S d^2 z f(z) G_2(z, x)$.

3. *The heat kernel.* In $\mathbb{R}^2$, with variables x and t , let D be the region $t > 0$, so S is the x -axis. We look for solutions to the *heat equation*

$$\frac{\partial f}{\partial t} - \frac{\partial^2 f}{\partial x^2} = 0,$$

with boundary value $f(x, 0) = f_0(x)$. Since $G(y, x, t) = \exp(-(x - y)^2/4t)/\sqrt{4\pi t}$ is a solution to (3) and approaches $\delta(x - y)$ as $t \rightarrow 0$, the solution to our problem is

$$f(x, t) = \int G(y, x, t) f_0(y) dy = \frac{1}{\sqrt{4\pi t}} \int \exp(-(x - y)^2/4t) f_0(y) dy.$$

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60

Linear Algebra in Biomolecular Modeling

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60.1 Introduction

Biomolecular modeling is an active research area in computational biology. It studies the structures and functions of biomolecules by using computer modeling and simulation [Sch03]. Proteins are an important class of biomolecules. They are encoded in genes and produced in cells through genetic translation. Proteins are life supporting (or sometimes, destructing) ingredients (Figure 60.1) and are indispensable for almost all biological processes [Boy99]. In order to understand the diverse biological functions of proteins, the knowledge of the three-dimensional structures of proteins is essential. Several structure determination techniques have been used, including x-ray crystallography, nuclear magnetic resonance spectroscopy (NMR), and homology modeling. They all require intensive mathematical computing, ranging from data analysis to model building [Cre93].

As in all other types of scientific computing, linear algebra is one of the most powerful mathematical tools for biological computing. Here we review several subjects in biomolecular modeling, where linear algebra has played a major role, including mapping from distances to coordinates in NMR structure determination (Section 60.2), solving the Procrustes problem for structural comparison (Section 60.3), exploiting the structure of the Karle–Hauptman matrix in protein x-ray crystallography (Section 60.4), computing the fast and slow modes of protein motions (Section 60.5), and solving the flux balancing equations in metabolic network simulation (Section 60.6). The last subject actually involves the modeling of a large biological system, something beyond conventional biomolecular modeling, yet of increased research interests in computational systems biology [Kit99].

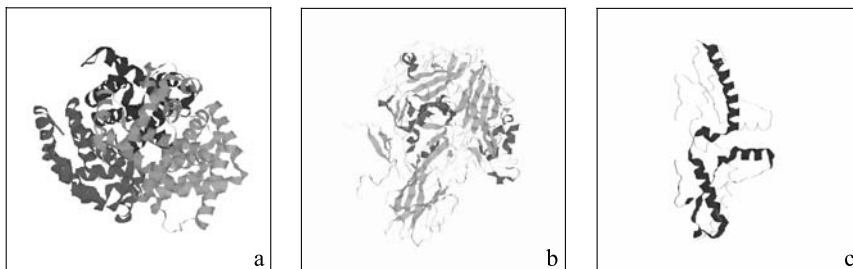


FIGURE 60.1 Example proteins: Humans have hundreds of thousands of different proteins (e.g., hemoglobin protein, 1BUW, in blood in 1a) and would not be able to maintain normal life even if short of a single type of protein. On the other hand, with the help of some proteins (e.g., protein, 2PLV, supporting poliovirus in 1b), viruses are able to grow, translate, integrate, and replicate, causing diseases. Some proteins themselves are toxic and even infectious such as the proteins in poisonous plants and in beef causing the Mad Cow Disease (e.g., prion protein, 1I4M-D, in human in 1c).

60.2 Mapping from Distances to Coordinates: NMR Protein Structure Determination

A fundamental problem in protein modeling is to find the three-dimensional structure of a protein and its relationship with the protein's biological function. One of the experimental techniques for structure determination is to use the nuclear magnetic resonance (NMR) to obtain some information on the distances for certain pairs of atoms in the protein and then find the coordinates of the atoms based on the obtained distance information. Mathematically, the second part of the work requires the solution of a so-called distance geometry problem, i.e., determine the coordinates for a set of points in a given topological space, given the distances for a subset of all pairs of points. We consider such a problem with the distances for all pairs of points assumed to be given.

Definitions:

The **coordinate vector** for atom i is a vector $\mathbf{x}_i = (x_{i,1}, x_{i,2}, x_{i,3})^T$, where $x_{i,1}$, $x_{i,2}$, and $x_{i,3}$ are the first, second, and third coordinates of atom i , respectively.

The **distance** between atoms i and j is defined as $d_{i,j} = \|\mathbf{x}_i - \mathbf{x}_j\|$, where $\mathbf{x}_i$ and $\mathbf{x}_j$ are coordinate vectors of atoms i and j , and $\|\cdot\|$ is the Euclidean norm.

The **coordinate matrix** for a protein is a matrix of coordinates denoted by $X = \{x_{i,j} : i = 1, \dots, n, j = 1, 2, 3\}$, where n is the total number of atoms in the protein, and row i of X is the coordinate vector of atom i .

The **distance matrix** for a protein is a matrix of distances denoted by $D = \{d_{i,j} : i, j = 1, \dots, n\}$, where $d_{i,j}$ is the distance between atoms i and j .

The problem of computing the coordinates of atoms (X) given a set of distances between pairs of atoms (D) is known as the **molecular distance geometry problem**.

Facts:

1. [Sax79] If the protein structure and, hence, X are known, D can immediately be computed from X . Conversely, if D is known or even partially known, X can also be obtained from D , but the computation is not as straightforward. The latter is proved to be *NP*-complete for arbitrary sparse distance matrices.

2. [Blu53] Choose a reference system so that the origin is located at the last atom, or in other words, $x_n = (0, 0, 0)^T$. Let X° be a submatrix of X , $X^\circ = \{x_{i,j} : i = 1, \dots, n-1, j = 1, 2, 3\}$, and D° be a matrix derived from D , $D^\circ = \{(d_{i,n}^2 - d_{i,j}^2 + d_{j,n}^2)/2 : i, j = 1, \dots, n-1\}$. Then, matrix D° is maximum rank 3 and $X^\circ X^{\circ T} = D^\circ$.
3. [CH88] Let $D^\circ = U\Sigma U^T$ be the singular-value decomposition of D° , where U is an orthogonal matrix and Σ a diagonal matrix with the singular values of D° along the diagonal. If D° is a matrix of rank less than or equal to 3, the decomposition can be obtained with U being $(n-1) \times 3$ and Σ being 3×3 , and $X^\circ = U\Sigma^{1/2}$ solves the equation $X^\circ X^{\circ T} = D^\circ$.

Algorithm 2: Computing Coordinates from Distances

Given an $n \times n$ distance matrix D ,

1. Compute $D^\circ = \{(d_{i,n}^2 - d_{i,j}^2 + d_{j,n}^2)/2 : i, j = 1, \dots, n-1\}$.
2. Decompose $D^\circ = U\Sigma U^T$ to obtain $X^\circ U[1:n-1, 1:3]\Sigma^{1/2}[1:3, 1:3]$.
3. $X[1:n-1, 1:3] = X^\circ[1:n-1, 1:3]$, $X[n, 1:3] = [0, 0, 0]$.

Examples:

1. Given the distances among four atoms, D , determine the coordinates of the atoms, X , where

$$D = \begin{bmatrix} 0 & \sqrt{2} & \sqrt{2} & 1 \\ \sqrt{2} & 0 & \sqrt{2} & 1 \\ \sqrt{2} & \sqrt{2} & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}.$$

Following Algorithm 1,

$$D^\circ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute the singular value decomposition of D° . Obviously, $D^\circ = U\Sigma U^T$, with

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \acute{O} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then,

$$X^\circ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

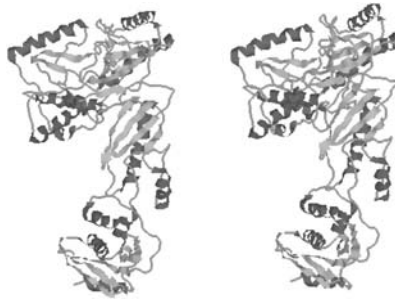


FIGURE 60.2 3D structures of protein 1HMV p66 subunit: The structure on the left was determined by x-ray crystallography, while on the right by solving a distance geometry problem given the distances for all the pairs of atoms. The RMSD for the two structures when compared on all the atoms is around $1.0\text{e-}04 \text{ \AA}$. (Photo courtesy of Qunfeng Dong.)

- Figure 60.2 shows two 3D structures of the p66 subunit of the HIV-1 retrotranscriptase (1HMV), one determined experimentally by x-ray crystallography [RGH95] and another computationally by solving a molecular distance geometry problem using the SVD method with the distance data generated from the known crystal structure. The RMSD (see description in section 60.3) for the two structures when compared on all the atoms is around $1.0\text{e-}04 \text{ \AA}$, showing that the two structures are almost identical.

60.3 The Procrustes Problem for Protein Structure Comparison

The structural differences between two proteins can be measured by the differences in the coordinates of the atoms for all corresponding atom pairs. The comparison is often required for either structural validation or functional analysis. The calculation can be done by solving a special linear algebra problem called the Procrustes problem [GL89].

Definitions:

Let X and Y be two $n \times 3$ coordinate matrices for two lists of atoms in proteins A and B , respectively, where $\mathbf{x}_i = (x_{i,1}, x_{i,2}, x_{i,3})^T$ is the coordinate vector of the i th atom selected from protein A to be compared with $\mathbf{y}_i = (y_{i,1}, y_{i,2}, y_{i,3})^T$, the coordinate vector of the i th atom selected from protein B . Assume that X and Y have been translated so that their centers of geometry are located at the same position, say, at the origin. Then, the structural difference between the two proteins can be measured by using **the root-mean-square deviation** (RMSD) of the structures, $\text{RMSD}(X, Y) = \min_Q \|X - YQ\|_F / \sqrt{n}$, where Q is a 3×3 rotation matrix and $QQ^T = I$, and $\|\cdot\|_F$ is the matrix Frobenius norm.

The RMSD is basically the smallest average coordinate errors of the structures for all possible rotations Q of structure Y to fit structure X . It is called **the Procrustes problem** for its analogy to the Greek story about cutting a person's legs to fit a fixed-sized iron bed. Note that X and Y may be the coordinate matrices for the same ($A = B$) or different ($A \neq B$) proteins and therefore, each pair of corresponding atoms do not have to be of the same type (when $A \neq B$). However, the number of atoms selected to compare must be the same from A and B ($\# \text{ rows of } X = \# \text{ rows of } Y$).

Facts:

- Let A and B be two matrices. Suppose that A is similar to B , then $\text{trace}(A) = \text{trace}(B)$. In particular, $\text{trace}(A) = \text{trace}(V^T AV)$, for any orthogonal matrix V .

2. [GL89] Let $C = Y^T X$ and $C = U \Sigma V^T$ be the singular-value decomposition of C . Then, $Q = UV^T$ minimizes $\|X - YQ\|_F$.

Algorithm 2: Computing the RMSD of Two Protein Structures

1. Compute the geometric centers of X and Y :

$$\mathbf{x}_c[j] = \left(\sum_{i=1}^n X[i, j] \right) / n, \quad j = 1, 2, 3;$$

$$\mathbf{y}_c[j] = \left(\sum_{i=1}^n Y[i, j] \right) / n, \quad j = 1, 2, 3.$$

2. Translate X and Y to the origin:

$$X = X - \mathbf{1}_n \mathbf{x}_c^T, \quad Y = Y - \mathbf{1}_n \mathbf{y}_c^T, \quad \mathbf{1}_n = (1, \dots, 1)^T \text{ in } \mathbb{R}^n.$$

3. Compute $C = Y^T X$ and $C = U \Sigma V^T$. Then,

$$Q = UV^T, \quad \text{RMSD}(X, Y) = \|X - YQ\|_F / \sqrt{n}.$$

Examples:

1. Suppose that X and Y are given as the following.

$$X = \begin{bmatrix} -1 & -1 & -2 \\ -1 & -1 & 0 \\ -1 & 1 & -2 \\ -1 & 1 & 0 \\ 1 & -1 & -2 \\ 1 & -1 & 0 \\ 1 & 1 & -2 \\ 1 & 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 1 & 2 \\ 1 & -1 & 2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{bmatrix}.$$

Then $\mathbf{x}_c = (0, 0, -1)^T$ and $\mathbf{y}_c = (0, 0, 1)^T$. Following the Step 2 in Algorithm 2, X and Y are changed to

$$X = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$Y = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix}.$$

Let $C = Y^T X$. Then,

$$C = \begin{bmatrix} 0 & -8 & 0 \\ 0 & 0 & -8 \\ 8 & 0 & 0 \end{bmatrix}.$$

Compute the singular value decomposition of C to obtain $C = U \Sigma V^T$, with

$$U = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}.$$

Then,

$$Q = UV^T = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}.$$

By calculating $\|X - YQ\|_F / \sqrt{n}$, we obtain $\text{RMSD}(X, Y) = 0$.

2. RMSD calculation has been widely used in structural computing. A straightforward application is for comparing and validating the structures obtained from different (x-ray crystallography, NMR, or homology modeling) sources for the same protein [Rho00]. Even from the same source, such as NMR, multiple structures are often obtained, and the average RMSD for the pairs of the multiple structures has been calculated as an indicator for the consistency and sometimes the flexibility of the structures [SNB03]. It has also been an important tool for structural classification, motif recognition, and structure prediction, where a large number of different proteins need to be aligned and compared [EJT00]. Figure 60.3 gives an example of using RMSD to compare NMR and x-ray crystal structures. Three structures of the second domain of the immunoglobulin-binding protein



FIGURE 60.3 NMR and x-ray crystal structures of 2IGG: Two NMR structures of 2IGG are superposed to its x-ray crystal structure to find out which one is closer to the x-ray crystal structure (dark line). The RMSD values for the two NMR structures against the x-ray structure are 1.97 Å and 1.75 Å, respectively. (Photo courtesy of Feng Cui.)

(2IGG) [LDS92] are displayed in the figure. Two of them are NMR structures. They are compared using RMSD against the x-ray structure (dark line).

60.4 The Karle–Hauptman Matrix in X-Ray Crystallographic Computing

X-ray crystallography has been a major experimental tool for protein structure determination and is responsible for about 80% of 30,000 protein structures so far determined and deposited in the Protein Data Bank [BWF00]. The structure determination process involves crystallizing the protein, applying x-ray to the protein crystal to obtain x-ray diffractions, and using the diffraction data to deduce the electron density distribution of the crystal (Figure 60.4). Once the electron density distribution of the crystal is known, a 3D structure for the protein can be assigned [Dre94].

Definitions:

Define $\rho: R^3 \rightarrow R$ to be **the electron density distribution function** for a protein and F_H in complex space C to be the structure factor representing the diffraction spot specified by the integral triplet H [Dre94].

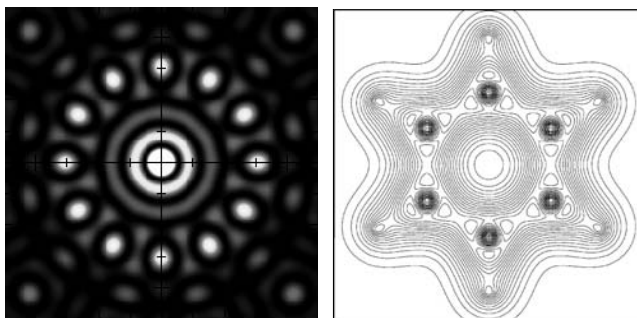


FIGURE 60.4 Example diffraction image and electron density map: The left one is the diffraction image of a 12-atom polygon generated by the program in [PNB01]. The right one is the electron density map of benzene generated by Stewart using program DENSITY in MOPAC [Ste02].

A **Karle–Hauptman matrix** for a set of structure factors $\{F_H : H = H_0, \dots, H_{n-1}\}$ is defined as

$$K = \begin{bmatrix} F_{H_0} & F_{H_{n-1}} & \cdots & F_{H_1} \\ F_{H_1} & F_{H_0} & \cdots & F_{H_2} \\ \vdots & \vdots & \ddots & \vdots \\ F_{H_{n-1}} & F_{H_{n-2}} & \cdots & F_{H_0} \end{bmatrix} \quad [KH52].$$

Facts:

1. [Dre94] The electron density distribution function ρ can be expanded as a Fourier series with the structure factors F_H as the coefficients. In other words, F_H is a Fourier transform of ρ .

$$\rho(r) = \sum_{H \in \mathbb{Z}^3} F_H \exp(-2\pi i H^T r),$$

$$F_H = \int_{\mathbb{R}^3} \rho(r) \exp(2\pi i H^T r) dr.$$

2. [Bri84] If K is a Karle–Hauptman matrix, then the inverse of K is also a Karle–Hauptman matrix and can be formed directly as

$$K^{-1} = \begin{bmatrix} E_{H_0} & E_{H_{n-1}} & \cdots & E_{H_1} \\ E_{H_1} & E_{H_0} & \cdots & E_{H_2} \\ \vdots & \vdots & \ddots & \vdots \\ E_{H_{n-1}} & E_{H_{n-2}} & \cdots & E_{H_0} \end{bmatrix},$$

where

$$E_{H_j} = \int_{\mathbb{R}^3} \rho^{-1}(r) \exp(2\pi i H_j^T r) dr, \quad j = 0, 1, \dots, n-1.$$

3. [GL89] Suppose that we have a linear system $K\mathbf{x} = \mathbf{h}$, where K is an $n \times n$ Karle–Hauptman matrix and $\mathbf{h}$ an n -dimensional complex vector. If a conventional method, such as Gaussian Elimination, is used, the solution of the system usually takes $O(n^3)$ floating-point operations, which is expensive if n is larger than 1000 and if the solution is also required multiple times.
4. [Loa92], [WPT01] Since each element in the inverse matrix can be obtained by doing a Fourier transform for the inverse of ρ and only n distinct elements in the first column are required to form the whole matrix, the calculations can be done in $O(n \log n)$ floating-point operations by using the Fast Fourier Transform.
5. [TML97], [WPT01] The matrix K^{-1} as well as K has only n distinct elements listed repeatedly in the columns of the matrix with each column having the elements in the previous column circulated by one element from top to the bottom and then bottom to the top. This type of matrix is called the circulant matrix. According to the discrete convolution theory, if h is the Fourier transform of t , then $K^{-1}h$ can be computed by doing a Fourier transform for $\rho^{-1} \cdot t$, where t can be obtained through an inverse Fourier transform for h and the product $\cdot$ is applied component-wise. Therefore, the whole computation for the solution of $K\mathbf{x} = \mathbf{h}$ can be done with at most $O(n \log n)$ floating-point operations.

Examples:

1. Let $\rho = [0.1250, 0.1250, 0.5000, 0.1250, 0.1250]$ be an electron density distribution. Then $\rho^{-1} = [8, 8, 2, 8, 8]$, and the Fourier transforms for ρ and ρ^{-1} are equal to

$$F = [0.2000, -0.0607 + 0.0441i, 0.0232 - 0.0713i, 0.0232 + 0.0713i, -0.0607 - 0.0441i] \text{ and}$$

$$F_{\text{inv}} = [6.8000, 0.9708 - 0.7053i, -0.3708 + 1.1413i, -0.3708 - 1.1413i, 0.9708 + 0.7053i],$$

respectively. And it is not hard to verify that the inverse of the Karle–Hauptman matrix $K(F)$ formed by using F is equal to the Karle–Hauptman matrix $K_{\text{inv}}(F_{\text{inv}})$ formed by using F_{inv} .

2. The Karle–Hauptman matrix is an important matrix in x-ray crystallography computing, named after two Nobel Laureates, chemist Jerold Karle and mathematician Herbert Hauptman, who received the Nobel Prize in chemistry in 1985 for their work on the phase problem in x-ray crystallography [HK53]. The Karle–Hauptman matrix is frequently used for computing the covariance of the structure factors [KH52] or the electron density distribution that maximizes the entropy of a crystal system [WPT01].

60.5 Calculation of Fast and Slow Modes of Protein Motions

In a reduced model for protein, a residue is represented by a point, in many cases, the position of the backbone atom C_α or the sidechain atom C_β in the residue, and a protein is considered as a sequence of such points connected with strings [HL92]. If the reduced model of a protein is known, a so-called contact map can be constructed to show how the residues in the protein interact with each other. The map is represented by a matrix with its i, j -entry equal to -1 if residues i and j are within, say $7\text{\AA}$ distance, and 0 otherwise. The contact matrix can be used to compare different proteins. Similar contact patterns often imply structural or functional similarities between proteins [MJ85]. When a protein reaches its equilibrium state, the residues in contact can be considered as a set of masses connected with springs. A simple energy function can also be defined for the protein using the contact matrix.

Definitions:

Suppose that a protein has n residues with n coordinate vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$. A **contact matrix** Γ for the protein in its equilibrium state can be defined such that

$$\tilde{A}_{i,j} = \begin{cases} -1, & ||\mathbf{x}_i - \mathbf{x}_j|| \leq 7 \text{ \AA} \\ 0, & \text{otherwise} \end{cases} \quad i \neq j = 1, \dots, n \quad [\text{HBE97}].$$

$$\tilde{A}_{i,i} = -\sum_{j=1}^n \tilde{A}_{i,j} \quad i = 1, \dots, n$$

A **potential energy function** E for a protein at its equilibrium state can be defined such that for any vector $\Delta \mathbf{x} = (\Delta \mathbf{x}_1, \dots, \Delta \mathbf{x}_n)^T$ of the displacements of the residues from their equilibrium positions,

$$E(\Delta \mathbf{x}) = \frac{1}{2} k \Delta \mathbf{x}^T \tilde{A} \Delta \mathbf{x},$$

where k is a spring constant.

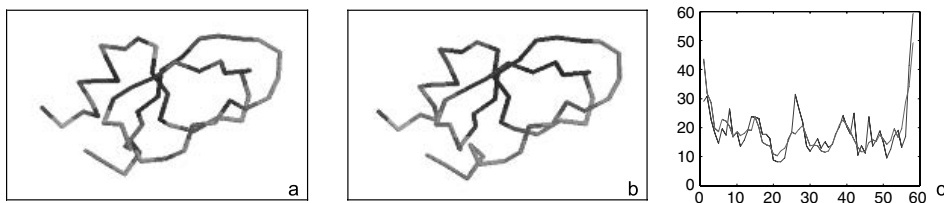


FIGURE 60.5 Mean-square fluctuations: The fluctuations for protein 2KNT based on the mean-square fluctuations calculated with GNM (a) and the B-factors determined by x-ray crystallography (b). The two sets of values show a high correlation (0.82) (c). (Photos courtesy of Di Wu.)

Facts:

1. [HBE97] Given a potential energy function E , the probability for a protein to have a displacement $\Delta \mathbf{x}$ at temperature T should be subject to the Boltzmann distribution,

$$p_T(\Delta \mathbf{x}) = \frac{1}{Z} \exp(-E(\Delta \mathbf{x})/k_B T) = \frac{1}{Z} \exp(-k \Delta \mathbf{x}^T \tilde{\mathbf{A}} \Delta \mathbf{x}/2k_B T),$$

where Z is the normalization factor and k_B the Boltzmann constant.

2. [HBE97] Let the singular-value decomposition of Γ be given as $\Gamma = U \Lambda U^T$. Then, the mean-square residue fluctuations of a protein at its equilibrium state can be estimated as

$$\langle \Delta \mathbf{x}_i, \Delta \mathbf{x}_i \rangle \equiv \frac{1}{Z} \int_{R^{3n}} \Delta \mathbf{x}_i^T \Delta \mathbf{x}_i \exp(-E(\Delta \mathbf{x})/k_B T) d\Delta \mathbf{x} = \sum_{j=1}^n k_B T U_{i,j} \tilde{\mathbf{E}}_{j,j}^{-1} U_{i,j} / k.$$

Examples:

1. The energy model defined above for a protein at its equilibrium state is called the Gaussian Network Model. The model can be used to find how the residues in the protein move around their equilibrium positions dynamically and in particular, to estimate the so-called mean-square fluctuations for the residues $\langle \Delta \mathbf{x}_i, \Delta \mathbf{x}_i \rangle$, $i = 1, \dots, n$. If the mean-square fluctuation is large, the residue is called hot, and otherwise, is cold, which often correlates with the experimentally detected average atomic fluctuation such as the B-factor in x-ray crystallography [Dre94] and the order parameter in NMR [Gun95]. In fact, the Gaussian Network Model is equivalent to the Normal Mode Analysis for predicting the mean-squares residue fluctuations of a protein, with the energy function defined for the residues instead of the atoms.
2. Figure 60.5 shows the mean-square fluctuations calculated using the Gaussian Network Model for the protein 2KNT and the comparison with the B-factors of the structure determined by x-ray crystallography. The two sets of values appear to be highly correlated. Based on the facts stated above, the calculation of the mean-squares fluctuations requires only a singular-value decomposition of the contact matrix for the protein.

60.6 Flux Balancing Equation in Metabolic Network Simulation

A metabolic system is maintained through constant reactions or interactions among a large number of biological and chemical compounds called metabolites [Fel97]. The reaction network describes the structure of a metabolic system and is key to the study of the metabolic function of the system. Figure 60.6 shows the reaction network for an example metabolic system of five metabolites given in [SLP00].

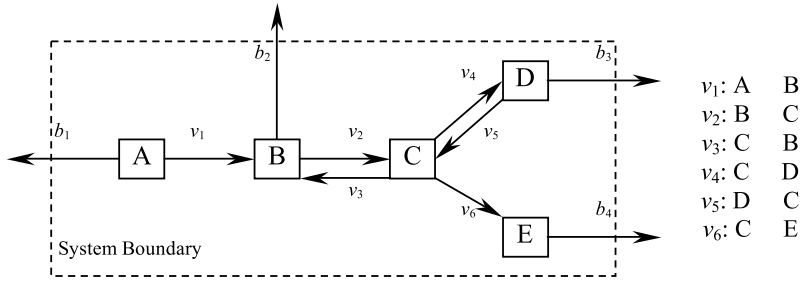


FIGURE 60.6 Example metabolic networks: A, B, C, D, E are metabolites; v_j , $j = 1, \dots, 6$ are internal fluxes; b_j , $j = 1, \dots, 4$ are external fluxes. Each flux v_j corresponds to an internal reaction.

Definitions:

Each metabolite has a concentration, which changes constantly. **The rate of the change** is proportional to the amount of the metabolite consumed or produced in all the reactions.

Let C_i be **the concentration of metabolite i** . Let v_j be **the chemical flux** in reaction j , i.e., the amount of metabolites produced in reaction j per mole. Then,

$$\frac{dC_i}{dt} = \sum_{j=1}^n s_{i,j} v_j,$$

where $s_{i,j}$ is the **stoichiometric coefficient of metabolite i** in reaction j , and $s_{i,j} = \pm k$, if $\pm k$ moles of metabolite i are produced (or consumed) in reaction j .

Let $C = (C_1, \dots, C_m)^T$ be a vector of concentrations of m metabolites, and $v = (v_1, \dots, v_n)^T$ a vector of fluxes of n reactions. Then, the equations can be written in a compact form,

$$\frac{dC}{dt} = S v,$$

where $S = \{s_{i,j} : i = 1, \dots, m, j = 1, \dots, n\}$ is called the **stoichiometry matrix**, and the equations are called **the reaction equations** [HS96].

The fluxes are functions of the concentrations and some other system parameters. Therefore, the above reaction equations are nonlinear equations of C . However, when the system reaches its equilibrium, $dC/dt = Sv = 0$, and the vector v becomes constant and is called a solution of **the steady-state flux equation** $Sv = 0$ [HS96].

The steady-state fluxes are important quantities for characterizing metabolic networks. They can be obtained by solving the steady-state flux equation $Sv = 0$. However, since the number of reactions is usually larger than the number of metabolites, the solution to the equation is not unique. Also, because the internal fluxes are nonnegative, the solution set forms a convex cone, called the **steady-state flux cone**.

Usually, a convex cone can be defined in terms of a set of extreme rays such that any vector in the cone can be expressed as a nonnegative linear combination of the extreme rays,

$$\text{cone}(S) = \{v = \sum_{i=1}^l w_i p_i, \quad w_i \geq 0\},$$

where $p = \{p_1, \dots, p_l\}$ is a set of extreme rays. An **extreme ray** is a vector that cannot be expressed as a nonnegative linear combination of any other vectors in the cone.

A set of vectors is said to be systematically independent if none of them can be expressed as a nonnegative linear combination of others. Since the extreme rays can be used to express all the vectors in a convex cone, they are also called the generating vectors of the cone. For metabolic networks, they are called the **extreme pathways** [PPP02].

Facts:

1. [PPP02] A convex flux cone has a set of systematically independent generating vectors or extreme pathways. They are unique up to positive scalar multiplications. If the extreme pathways of the convex flux cone of a metabolic network are found, all the solutions for the steady-state flux equation can be generated by using the extreme pathways. In other words, the extreme pathways provide a unique description for the solution space of the steady-state flux equation, and can be used to characterize the whole steady-state capacity of the system.
2. [PPP02] Let P be a matrix with each column corresponding to an extreme pathway of a given metabolic network. Let Q be the binary form of P such that $Q_{i,j} = 1$ if $P_{i,j} \neq 0$ and $Q_{i,j} = 0$ otherwise. Then, the diagonal elements of $Q^T Q$ are equal to the lengths of the extreme pathways, while the diagonal elements of $Q Q^T$ show the numbers of extreme pathways the reactions participate in.

Examples:

1. Consider the example network in Figure 60.6 and let S be the stoichiometric matrix with 10 columns for the internal ($v_1, \dots, v_6$) as well as external ($b_1, \dots, b_4$) fluxes,

$$S = \begin{array}{cccccccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & b_1 & b_2 & b_3 & b_4 \\ \left[\begin{array}{cccccccccc} -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \end{array} \right] & \begin{array}{l} \leftarrow A \\ \leftarrow B \\ \leftarrow C \\ \leftarrow D \\ \leftarrow E \end{array} \end{array}$$

Then, by using an appropriate algorithm (such as the one given in [SLP00]), a matrix of 7 extreme pathways of the system can be found as follows:

$$P^T = \begin{array}{cccccccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & b_1 & b_2 & b_3 & b_4 \\ \left[\begin{array}{cccccccccc} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & -1 & 1 \end{array} \right] & \begin{array}{l} \leftarrow p_1 \\ \leftarrow p_2 \\ \leftarrow p_3 \\ \leftarrow p_4 \\ \leftarrow p_5 \\ \leftarrow p_6 \\ \leftarrow p_7 \end{array} \end{array},$$

where row i corresponds to extreme pathway p_i , $i = 1, \dots, 7$.

2. By forming the binary form Q for P and computing

$$Q^T Q = \begin{bmatrix} 3 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 4 & 2 & 2 & 1 & 1 \\ 1 & 1 & 2 & 4 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 4 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 2 & 1 & 4 \end{bmatrix}$$

and

$$QQ^T = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 3 & 1 & 1 & 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 3 & 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & 0 & 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 1 & 1 & 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 & 1 & 0 & 2 & 3 & 1 \\ 0 & 1 & 0 & 0 & 1 & 2 & 0 & 1 & 1 & 2 \end{bmatrix},$$

we obtain, from the diagonal elements of the matrices, the lengths of the pathways, $p_1: 3$, $p_2: 2$, $p_3: 4$, $p_4: 4$, $p_5: 4$, $p_6: 2$, $p_7: 4$, and the participations of the reactions in the extreme pathways, $v_1: 1$, $v_2: 3$, $v_3: 2$, $v_4: 2$, $v_5: 3$, $v_6: 2$, $b_1: 1$, $b_2: 4$, $b_3: 3$, $b_4: 2$. The off-diagonal elements of $Q^T Q$ show the numbers of common reactions in different extreme pathways. For example, $(Q^T Q)_{3,4} = 2$ means that p_3 and p_4 share two common reactions, and $(Q^T Q)_{3,5} = 1$ means that p_3 and p_5 have one common reaction. The off-diagonal elements of QQ^T show the numbers of extreme pathways in which different reactions participate. For example, $(QQ^T)_{2,3} = 1$ means that v_2 and v_3 participate in one extreme pathway together, and $(QQ^T)_{2,8} = 2$ means that v_2 and b_2 participate in two extreme pathways together.

60.7 Conclusion

In this chapter, we have reviewed several subjects in biomolecular modeling, where linear algebra has played a central role in related computations. The review has focused on simple showcases and demonstrated important applications of linear algebra in biomolecular modeling. The subjects discussed are of great research interest in computational biology and are related directly to the solutions of many critical but challenging computational problems in biology yet to be solved, including the general distance geometry problem in NMR, the phase problem in x-ray crystallography, the structural comparison problem in comparative modeling, molecular dynamics simulation, and biosystems modeling and optimization, which we have not elaborated in detail here, but the interested readers can further explore.

Linear algebra has also been used to support many basic algebraic calculations required for solving other types of mathematical problems in biomolecular modeling, such as the optimization problems in potential energy minimization [WS99], the initial value problem in molecular dynamics simulation [BKP88], and the boundary value problem in simulation of protein conformational transformation [EMO99]. They are usually straightforward or routine linear algebra calculations, so we have not covered them in this chapter, but they should be considered as equally important as the applications we have discussed.

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61

Coding Theory

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Sometimes errors occur when data is transmitted over a channel. A binary 0 may be corrupted to a 1, or vice versa. Error control coding adds redundancy to a transmitted message allowing for detection and correction of errors. For example, if 0 stands for “no” and 1 stands for “yes”, a single bit error will completely change a message. If we repeat the message seven times, so 0000000 stands for “no” and 1111111 stands for “yes”, then it would require at least 4 bit errors to change a message, assuming that a message is decoded to either 0000000 or 1111111 according to which bit is in the majority in the received string. Thus, this simple code can be used to correct up to 3 errors. This code also detects up to 6 errors, in the sense that if between 1 and 6 errors occur in transmission, the received string will not be a codeword, and the receiver will recognize that one or more errors have occurred. The extra bits provide *error protection*.

Let $\mathbb{F} = \mathbb{F}_q$ be the finite field with q elements. It is customary in the coding literature to write message blocks and vectors of the vector spaces $\mathbb{F}^k$ and $\mathbb{F}^n$ as row vectors, and we shall follow that practice in this chapter.

61.1 Basic Concepts

Definitions:

A **block code of length n over $\mathbb{F}$** is a subset $\mathcal{C}$ of the vector space $\mathbb{F}^n$. Elements of $\mathcal{C}$ are **codewords**. Elements of the finite field $\mathbb{F}$ form the **alphabet**.

Message blocks of length k are strings of k symbols from the alphabet. Message blocks may also be identified with vectors in the vector space $\mathbb{F}^k$.

An **encoder** is a one-to-one mapping

$$\varphi : \mathbb{F}^k \longrightarrow \mathbb{F}^n$$

whose image is contained in the code $\mathcal{C}$.

The process of applying φ to a k -digit message block to obtain a codeword is **encoding**.

A **noisy channel** is a channel in which transmission errors may occur.

A codeword $\mathbf{v} \in \mathcal{C} \subseteq \mathbb{F}^n$ that has been transmitted may be received as $\mathbf{y} = \mathbf{v} + \mathbf{e}$, where $\mathbf{e} \in \mathbb{F}^n$ is an **error vector**.

Given a received word $\mathbf{y} = \mathbf{v} + \mathbf{e} \in \mathbb{F}^n$, the receiver estimates the corresponding codeword, thus, obtaining $\tilde{\mathbf{v}} \in \mathcal{C}$. This estimation process is called **decoding**. If $\tilde{\mathbf{v}} = \mathbf{v}$, then correct decoding has occurred.

Decoding may be thought of as a mapping $\psi : \mathbb{F}^n \rightarrow \mathcal{C}$. The mapping ψ is a **decoder**. Generally a decoder ψ should have the property that if the received word is in fact a codeword, it is decoded to itself.

The **word error rate** for a particular decoder is the probability P_{err} that the decoder outputs a wrong codeword.

The process of encoding, transmission, and decoding is summarized by

$$\begin{array}{ccccccc} \mathbb{F}^k & \xrightarrow{\varphi} & \mathbb{F}^n & \xrightarrow{\text{n.t.}} & \mathbb{F}^n & \xrightarrow{\psi} & \mathcal{C} \\ \mathbf{u} & \mapsto & \mathbf{v} & \mapsto & \mathbf{y} & \mapsto & \tilde{\mathbf{v}}, \end{array}$$

where “n.t.” stands for “noisy transmission.”

For $\mathbf{v}, \mathbf{w} \in \mathbb{F}^n$ the **Hamming distance**, or simply the **distance**, between $\mathbf{v}$ and $\mathbf{w}$ is $d(\mathbf{v}, \mathbf{w}) = |\{i \mid v_i \neq w_i\}|$.

The **distance of a code** $\mathcal{C} \subseteq \mathbb{F}^n$ is defined by $d(\mathcal{C}) = \min\{d(\mathbf{v}, \mathbf{w}) \mid \mathbf{v}, \mathbf{w} \in \mathcal{C} \text{ and } \mathbf{v} \neq \mathbf{w}\}$.

The **Hamming weight** or **weight** of $\mathbf{v}$ is defined to be $\text{wt}(\mathbf{v}) = d(\mathbf{v}, \mathbf{0})$.

The **closed ball of radius t about the vector $\mathbf{v} \in \mathbb{F}^n$** is the set

$$B(\mathbf{v}, t) = \{\mathbf{y} \in \mathbb{F}^n \mid d(\mathbf{v}, \mathbf{y}) \leq t\}.$$

Nearest-neighbor decoding refers to decoding a received word $\mathbf{y}$ to a nearest codeword (with respect to Hamming distance).

A code $\mathcal{C} \subseteq \mathbb{F}^n$ is called a **perfect code** if there exists a nonnegative integer t such that

1. $\mathbb{F}^n = \bigcup_{\mathbf{v} \in \mathcal{C}} B(\mathbf{v}, t)$, and
2. For distinct $\mathbf{v}, \mathbf{w} \in \mathcal{C}$, $B(\mathbf{v}, t) \cap B(\mathbf{w}, t) = \emptyset$.

That is, $\mathbb{F}^n$ is the disjoint union of closed t -balls about the codewords.

Facts:

1. The minimum number of symbol errors it would take for $\mathbf{v}$ to be changed to $\mathbf{w}$ is $d(\mathbf{v}, \mathbf{w})$.
2. A nearest-neighbor decoder need not be unique as it is possible that some received words $\mathbf{y}$ might have two or more codewords at an equal minimum distance. The decoder may decode $\mathbf{y}$ to any of these nearest codewords.
3. [Hil86, p. 5] The Hamming distance is a metric on $\mathbb{F}^n$.
4. [Hil86, p. 7] If the distance of a code $\mathcal{C}$ is d , then $\mathcal{C}$ can detect up to $d - 1$ errors. It can correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors.

Examples:

1. The binary message block 1011 can be identified with the vector $(1, 0, 1, 1) \in \mathbb{F}_2^4$.
2. The distance between the binary strings 0011 and 1110 is 3 since these two bit strings differ in three places.
3. We may picture nearest-neighbor decoding by taking a closed ball of expanding radius about the received word $\mathbf{y}$. As soon as the ball includes a codeword, that codeword is a nearest-neighbor to which we may decode $\mathbf{y}$.

61.2 Linear Block Codes

Definitions:

A **linear block code of length n and dimension k** over $\mathbb{F} = \mathbb{F}_q$ is a k -dimensional *linear* subspace, $\mathcal{C}$, of the vector space $\mathbb{F}^n$. One says $\mathcal{C}$ is an $[n, k]$ -code over $\mathbb{F}$. If one wants to include the distance as a parameter, one says that $\mathcal{C}$ is an $[n, k, d]$ -code over $\mathbb{F}$. A **binary linear block code (BLBC)** is a linear block code over the binary field $\mathbb{F}_2 = \{0, 1\}$.

Two linear codes over $\mathbb{F}_q$ are **equivalent** if one can be obtained from the other by steps of the following types: (i) permutation of the coordinates of the codewords, and (ii) multiplication of the field elements in one coordinate of the code by a nonzero scalar in $\mathbb{F}_q$.

The **rate** of a linear $[n, k]$ -code is $\frac{k}{n}$.

Let $\mathcal{C}$ be a linear $[n, k]$ -code over $\mathbb{F}$. A **generator matrix for $\mathcal{C}$** is a matrix $G \in \mathbb{F}^{k \times n}$ whose k rows form a basis for $\mathcal{C}$. A generator matrix for $\mathcal{C}$ is also called an **encoder for $\mathcal{C}$** .

A generator matrix G for an $[n, k]$ -code is a **systematic encoder** if G can be partitioned, perhaps after a column permutation, as $G = [I_k \mid A]$, where $A \in \mathbb{F}^{k \times (n-k)}$. Then $\mathbf{u} \in \mathbb{F}^k$ is encoded as $\mathbf{u}G = [\mathbf{u} \mid \mathbf{u}A]$ so that the first k symbols of the codeword $\mathbf{u}G$ are the message word $\mathbf{u}$, and the last $n - k$ symbols form the **parity check** symbols.

Let $\mathcal{C}$ be a linear $[n, k]$ -code over $\mathbb{F}$. A **parity check matrix for $\mathcal{C}$** is a matrix $H \in \mathbb{F}^{(n-k) \times n}$ such that $\mathcal{C} = \{\mathbf{v} \in \mathbb{F}^n \mid \mathbf{v}H^T = \mathbf{0} \in \mathbb{F}^{(n-k)}\}$.

Let $\mathcal{C}$ be a linear $[n, k]$ -code over $\mathbb{F}$ with generator matrix $G \in \mathbb{F}^{k \times n}$ and parity check matrix $H \in \mathbb{F}^{(n-k) \times n}$. Then H itself is the generator matrix of another code, denoted $\mathcal{C}^\perp$. $\mathcal{C}^\perp$ is the **dual code** of $\mathcal{C}$.

A linear block code, $\mathcal{C}$, is **self-dual** if $\mathcal{C} = \mathcal{C}^\perp$.

Let $\mathcal{C}$ be an $[n, k]$ -code with parity check matrix H . For any vector $\mathbf{y} \in \mathbb{F}^n$, the **syndrome of $\mathbf{y}$** is the vector $\mathbf{y}H^T \in \mathbb{F}^{(n-k)}$.

Facts:

1. [Hil86, p. 47] There are q^k codewords in an $[n, k]$ -code over $\mathbb{F} = \mathbb{F}_q$.
2. The rate, $\frac{k}{n}$, of an $[n, k]$ -code tells what fraction of transmitted symbols carry information. The complementary fraction, $\frac{n-k}{n}$, tells how much of the transmission is redundant (for error protection).
3. If G is a generator matrix for the linear $[n, k]$ -code, $\mathcal{C}$, then $\mathcal{C} = \{\mathbf{u}G \mid \mathbf{u} \in \mathbb{F}^k\}$. Thus, the mapping $\varphi : \mathbb{F}^k \rightarrow \mathbb{F}^n$ given by $\varphi(\mathbf{u}) = \mathbf{u}G$ is an encoder for $\mathcal{C}$.
4. [Hil86, p. 48] The distance of a linear code is equal to the minimum weight of all nonzero codewords.
5. Every $[n, k]$ linear block code has an $(n - k) \times n$ parity check matrix.
6. For a given $[n, k]$ linear code $\mathcal{C}$ with generator matrix G , an $(n - k) \times n$ matrix H is a parity check matrix for $\mathcal{C}$ if and only if the rows of H are a basis for the left kernel of G^T .
7. If H is a parity check matrix for the linear code $\mathcal{C}$, then $\mathcal{C}$ is the left kernel of H^T .
8. A generator matrix of $\mathcal{C}$ is a parity check matrix of $\mathcal{C}^\perp$.
9. [MS77, p. 33] Suppose $H \in \mathbb{F}^{(n-k) \times n}$ is a parity check matrix for an $[n, k]$ -code $\mathcal{C}$ over $\mathbb{F}$. Then the distance of $\mathcal{C}$ is equal to the smallest number of columns of H for which a linear dependency may be found.
10. [Hil86, p. 70] If $G = [I_k \mid A]$ is a systematic encoder for the $[n, k]$ -code $\mathcal{C}$, then $H = [-A^T \mid I_{n-k}]$ is a parity check matrix for $\mathcal{C}$.
11. [MS77, p. 22] *Shannon's Coding Theorem for the Binary Symmetric Channel*. Assume that the transmission channel has a fixed symbol error probability p that the symbol 1 is corrupted to a 0 or vice versa. Let

$$C := 1 + p \log_2 p + (1 - p) \log_2 (1 - p).$$

Then for every $\epsilon > 0$ and every $R < C$ there exists, for any sufficiently large n , a binary linear $[n, k]$ code having transmission rate $k/n \geq R$ and word error probability $P_{\text{err}} \leq \epsilon$.

Examples:

1. The binary repetition code of length 7 is a $[7, 1]$ BLBC. A generator matrix is

$$G = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

This code consists of two codewords, namely 0000000 and 1111111. The rate of this code is $\frac{1}{7}$. For every 7 bits transmitted, there is only 1 bit of information, with the remaining 6 redundant bits providing error protection. The distance of this code is 7. A parity check matrix for this code is the 6×7 matrix

$$H = [J_{6,1} \mid I_6].$$

2. *Binary even weight codes.* The binary even weight code of length n is a $[n, n-1]$ -code that consists of all even weight words in $\mathbb{F}_2^n$. A generator matrix is given by

$$G = [I_{n-1} \mid J_{(n-1),1}].$$

This code detects any single error, but cannot correct any errors. A binary even weight code may also be constructed by taking all binary strings of length $n-1$ and to each adding an n th parity check bit chosen so that the weight of the resulting string is even. The 8-bit ASCII code (American Standard Code for Information Interchange) is constructed in this manner.

Note that the binary even weight code of length n and the binary repetition code of length n are dual codes.

3. The ISBN (International Standard Book Number) code is a $[10, 9]$ -code over $\mathbb{F}_{11}$ (the finite field $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, X\}$ where field operations are performed modulo 11. X is the symbol for 10 in this field). This code may be defined by the parity check matrix

$$H = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & X \end{bmatrix}.$$

Thus, a string of 10 digits, $v_1 v_2 v_3 v_4 v_5 v_6 v_7 v_8 v_9 v_{10}$, from $\mathbb{F}_{11}$ is a valid ISBN if and only if $\sum_{i=1}^{10} i v_i \equiv 0 \pmod{11}$. The ISBN code will detect any single error or any error where two digits are interchanged (transposition error). The first nine digits of an ISBN carry information about the book (such as publisher). The tenth digit is a check digit. Given the first 9 digits of an ISBN, the check digit may be computed by forming the sum $\sum_{i=1}^9 i v_i$ and reducing modulo 11. The value thus obtained is v_{10} (if the value is 10, then $v_{10} = X$).

Note: Currently the ISBN system does not use X as any of the first 9 digits. However, mathematically there is no reason not to use X anywhere in an ISBN.

4. The binary linear $[8, 4]$ code with generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

is a self-dual code (note that over the binary field, $GG^T = 0$). This code is called the extended binary $[8, 4, 4]$ code. All pairs of different codewords are at distance 4 or 8.

61.3 Main Linear Coding Problem and Distance Bounds

Definitions:

Let $A_q(n, d)$ be the largest possible number so that there exists a block code $\mathcal{C} \subseteq \mathbb{F}_q^n$ of distance $d = d(\mathcal{C})$ and cardinality $|\mathcal{C}| = A_q(n, d)$. The **Main Coding Theory Problem** asks for the determination of $A_q(n, d)$ for different values of n, d, q .

Similarly, let $B_q(n, d)$ be the largest possible number so that there exists a linear $[n, k]$ block code $\mathcal{C}$ of distance $d = d(\mathcal{C})$ and cardinality $|\mathcal{C}| = B_q(n, d)$. The **Main Linear Coding Problem** asks for the determination of $B_q(n, d)$ for different values of n, d, q .

An $[n, k, d]$ linear code is **maximum distance separable (MDS)** if $d = n - k + 1$.

Facts:

1. Finding a value of $B_q(n, d)$ is equivalent to finding the largest dimension, k , for which there is an $[n, k, d]$ -code over $\mathbb{F}_q$. For such maximal k , we have $B_q(n, d) = q^k$.
2. $B_q(n, d) \leq A_q(n, d)$ for all n, q, d .
3. [Rom92, p. 170] $A_q(n, 1) = B_q(n, 1) = q^n$.
4. $B_q(n, 2) = q^{n-1}$. Such a code is achieved by taking $\mathcal{C}$ to be a code with parity check matrix $J_{1,n}$.
5. [Rom92, p. 170] $A_q(n, n) = B_q(n, n) = q$.
6. [Rom92, p. 170] For $n \geq 2$, $A_q(n, d) \leq q A_q(n-1, d)$.
7. [Hil86, Theorem 14.5] $B_q(n, 3) = q^{n-r}$, where r is the unique integer such that

$$(q^{r-1} - 1)/(q - 1) < n \leq (q^r - 1)/(q - 1).$$

8. [Rom92, Theorem 4.5.7] *The sphere-packing bound.* Suppose $\mathcal{C}$ is a block code of length n over $\mathbb{F}_q$ with distance d . Let $t = \lfloor \frac{d-1}{2} \rfloor$. Then

$$A_q(n, d) \left(1 + \binom{n}{1}(q-1) + \binom{n}{2}(q-1)^2 + \cdots + \binom{n}{t}(q-1)^t \right) \leq q^n.$$

9. [Hil86, pp. 20–21] The code $\mathcal{C}$ is perfect if and only if equality holds in the sphere-packing bound.
10. [Rom92, Theorem 4.5.7] Suppose $\mathcal{C}$ is a block code of length n over $\mathbb{F}_q$ with distance d . Then

$$A_q(n, d) \leq q^{n-d+1}.$$

11. *The Singleton bound for linear block codes.* For any $[n, k, d]$ linear code over $\mathbb{F}_q$, we have

$$d \leq n - k + 1.$$

12. An $(n - k) \times n$ matrix H with entries in $\mathbb{F}$ is the parity check matrix of an MDS code if and only if any $n - k$ columns of H are linearly independent. Equivalently, H has the property that any full size minor of H is invertible.
13. [MS77, pp. 546–547] *The Griesmer bound.* Let $N(k, d)$ be the length of the shortest binary linear code of dimension k and distance d . Then one has

$$N(k, d) \geq \sum_{i=0}^{k-1} \left\lceil \frac{d}{2^i} \right\rceil.$$

The right-hand side in the above inequality is, of course, always at least $d + k - 1$.

14. [Rom92, Theorem 4.5.16] *The Plotkin bound.* If $d \geq \frac{qn-n}{q}$, then

$$A_q(n, d) \leq \frac{dq}{d - qn - n}.$$

15. [Hil86, pp. 91–92] *The Gilbert Varshamov bound.* Assume q is a prime power. For any n and d , if k is the largest integer satisfying

$$q^k < \frac{q^n}{\sum_{i=0}^{d-2} \binom{n-1}{i} (q-1)^i},$$

then $A_q(n, d) \geq q^k$.

Examples:

1. Consider the binary $[7,4]$ code $\mathcal{C}$ defined by the parity check matrix

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

This code has $2^4 = 16$ elements. Since any two columns of H are linearly independent, but the first three columns are dependent, it follows that $d(\mathcal{C}) = 3$. So this code meets the sphere packing bound with $t = 1$. Observe that reading downward, the 7 columns of H form the numbers 1 through 7 in binary. This leads to a very nice nearest-neighbor decoding scheme that corrects any single error. If the received word is $\mathbf{y}$, the syndrome $\mathbf{s} = \mathbf{y}H^T$ consists of 3 bits. Interpret these 3 bits as a binary number, x . If x is 0, the received word is a codeword and we assume that no errors have occurred. If x is not 0, it is one of the numbers from 1 to 7. In $\mathbf{y}$ we change the bit in position x . This will give us a codeword (the unique codeword) at a distance of 1 from $\mathbf{y}$. We presume that this codeword is the intended transmission. For instance, if we receive the vector $\mathbf{y} = 1100100$, we compute the syndrome $\mathbf{s} = \mathbf{y}H^T = 110$. Interpreting 110 as a binary number, we obtain $x = 6$. We change the sixth bit of $\mathbf{y}$ and obtain the codeword 1100110. This codeword is at a Hamming distance 1 from $\mathbf{y}$, and we assume it was the intended transmission.

2. Assume $\alpha_1, \dots, \alpha_n$ are pairwise different nonzero elements of a finite field $\mathbb{F}$. Let

$$H := \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{n-k} & \alpha_2^{n-k} & \dots & \alpha_n^{n-k} \end{bmatrix}.$$

Then any full-size minor is the determinant of a Vandermonde matrix and, hence, nonzero. So H represents the parity check matrix of an MDS code.

61.4 Important Classes of Linear Codes 1

Definitions:

Let $\mathbb{F} = \mathbb{F}_q$.

Let r be a positive integer. Let $\mathbb{P}(\mathbb{F}^r)$ be the set of all 1-dimensional subspaces of $\mathbb{F}^r$. Choose a basis vector for each element of $\mathbb{P}(\mathbb{F}^r)$. Let H be a matrix whose columns are these basis vectors. The matrix H is the parity check matrix of a **Hamming code**, $\mathcal{C}$.

A **cyclic code** is a linear code $\mathcal{C}$ for which every cyclic shift of a codeword is again a codeword.

Let $\mathbb{F}[x]$ be the polynomial ring in one indeterminate over $\mathbb{F}$. Let $\langle x^n - 1 \rangle$ be the ideal generated by $x^n - 1$. The quotient ring $\mathbb{F}[x]/\langle x^n - 1 \rangle$ is denoted R_n . The polynomials of R_n are identified with vectors

in $\mathbb{F}^n$ via

$$a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} \mapsto [a_0 \ a_1 \ a_2 \ \cdots \ a_{n-1}].$$

Suppose $k < n$ are positive integers and $r = n - k$. A polynomial $g(x) = a_0 + a_1x + a_2x^2 + \cdots + a_rx^r$ dividing $x^n - 1 \in \mathbb{F}[x]$ is a **generator polynomial** of a cyclic $[n, k]$ -code, $\mathcal{C}$. The polynomial $h(x) = b_0 + b_1x + b_2x^2 + \cdots + b_kx^k$ chosen so that $g(x)h(x) = x^n - 1$ is the **parity check polynomial** of $\mathcal{C}$.

Facts:

1. The ring R_n forms an n -dimensional vector space over $\mathbb{F}$. The monomials $\{1, x, x^2, \dots, x^{n-1}\}$ form a basis. It further has an algebra structure over $\mathbb{F}$ (i.e., there is a well-defined multiplication). In fact, polynomials in R_n are multiplied as usual, but $x^n = 1$ in R_n , so exponents are reduced modulo n .
2. The cardinality of $\mathbb{P}(\mathbb{F}^r)$ is $n = (q^r - 1)/(q - 1)$. Thus, the parity check matrix, H , of a Hamming code is of size $r \times n$. The corresponding Hamming code $\mathcal{C}$ has q^k elements where $k = n - r$. Furthermore, the distance $d(\mathcal{C}) = 3$. The Hamming code reaches the sphere packing bound and hence is a perfect code in all cases.
3. [Hil86, pp. 147–148] For cyclic codes, R_n has the structure of a principal ideal ring (any ideal is generated by a single polynomial). Thus, any cyclic code has a generator polynomial $g(x)$ where $g(x)$ is a divisor of $x^n - 1$. Moreover, every cyclic code has a parity check polynomial, namely $h(x) = (x^n - 1)/g(x)$.
4. [Hil86, Chap. 12] If $\mathcal{C}$ is a cyclic code with generator polynomial $g(x) = a_0 + a_1x + a_2x^2 + \cdots + a_rx^r$ and parity check polynomial $h(x) = b_0 + b_1x + b_2x^2 + \cdots + b_kx^k$, then the corresponding generator matrix, G , and parity check matrix, H , of $\mathcal{C}$ are given by

$$G = \begin{bmatrix} a_0 & a_1 & a_2 & \cdots & a_r & 0 & 0 & \cdots & 0 \\ 0 & a_0 & a_1 & a_2 & \cdots & a_r & 0 & \cdots & 0 \\ 0 & 0 & a_0 & a_1 & a_2 & \cdots & a_r & \cdots & 0 \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & \cdots & a_0 & a_1 & a_2 & \cdots & a_r \end{bmatrix}$$

and

$$H = \begin{bmatrix} b_k & \cdots & b_2 & b_1 & b_0 & 0 & 0 & \cdots & 0 \\ 0 & b_k & \cdots & b_2 & b_1 & b_0 & 0 & \cdots & 0 \\ 0 & 0 & b_k & \cdots & b_2 & b_1 & b_0 & \cdots & 0 \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & \cdots & b_k & \cdots & b_2 & b_1 & b_0 \end{bmatrix}.$$

Examples:

1. For $q = 5$ and $r = 2$, we get the $[6, 4, 3]$ -Hamming code over $\mathbb{F}_5$ with parity check matrix

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 3 & 4 \end{bmatrix}$$